
Internship Report

Formalization, identification and decomposition of pluripotents Boolean networks

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Introduction

This report builds on foundational studies conducted by the TIMC laboratory concerning Boolean networks. The objective is to explore the existence of a larger pluripotent system which simulates multiple biological behaviors, emerging from the combination of individual biological behaviors found in distinct systems during evolutionary processes. In other words, we search for n -tuples of pluripotent networks $(R_1, R_2, \dots, R_m, R)$ of the same type, where their corresponding transition graphs are $(G_1, G_2, \dots, G_m, G)$, with each G_i being disjoint from the others, and G being a composite of all G_i .

In this context, we limit the number of sub-networks to $m = 2$ and specify the type of network as **Sign Boolean Networks**. However, this can be extended to a larger number of networks if one or more G_i can be continuously composed. To create a more flexible framework, it is feasible to relax the SBN Constraints. For example, we could implement only one SBN (either (G^0, R^0) or (G^1, R^1)) or use other types of Boolean Network such as **Threshold Boolean Networks**, **Propositional Boolean Networks**, etc.

The content is organized into four sections. The first one introduces some basic definitions, an efficient representation, and formally describes the objective. The second part explores the constraints and properties of weight matrices, offering proofs and examples, and introduces a composition algorithm for networks with a single control bit. The third part applies the same approach to networks with two control bits. (Note: due to the limited duration of the internship, this part may contain errors and still has room for optimization.) The appendix includes all possible weight matrices from 0D to 2D.

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Part I

Definitions

Chapter 1

Some basic notations

Let (R, G) be a tuple, where R is a pluripotent **Sign Boolean Network** (SBN) containing n nodes, and G is the **Transition Graph** (TG) in dimension n that corresponds to R , with 2^n vertices. Denote by $N_i, \forall i \in \llbracket 1, n \rrbracket$ the i -th node of R .

Definition 1.1 (Boolean variable x_i). $\forall i \in \llbracket 1, n \rrbracket$, the Boolean variable x_i represents the value of the node N_i .

Definition 1.2 (Weight Matrix of SBN R). The weight matrix W of a SBN R is defined as

$$W = \begin{pmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{12} & w_{22} & \cdots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1} & w_{n2} & \cdots & w_{nn} \end{pmatrix}$$

where $w_{ij} \in \mathbb{Z}$ represents the weight of edge from N_i to N_j , $\forall (i, j) \in \llbracket 1, n \rrbracket^2$.

Definition 1.3 (Sign function). We define the sign function as follows:

$$\begin{aligned} \text{sign} : \mathbb{Z}^n &\rightarrow \mathcal{B}^n = \{0, 1\}^n \\ \mathbf{t} &\mapsto \mathbb{1}_{(0, +\infty)}(\mathbf{t}) = \begin{pmatrix} \mathbb{1}_{(0, +\infty)}(t_1) \\ \vdots \\ \mathbb{1}_{(0, +\infty)}(t_i) \\ \vdots \\ \mathbb{1}_{(0, +\infty)}(t_n) \end{pmatrix} \end{aligned}$$

where $\mathbb{1}_{(0, +\infty)}$ is the indicator function

$$\mathbb{1}_{(0, +\infty)}(t_i) = \begin{cases} 1 & t_i > 0 \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in \llbracket 1, n \rrbracket$$

Remark 1.3.1. Note that $\text{sign} \circ \dots \circ \text{sign}(x) = \text{sign}(x)$.

Definition 1.4 (Function f_i). $\forall i \in \llbracket 1, n \rrbracket$, we define function f_i such that

$$\begin{aligned} f_i: \mathcal{B}^n &\rightarrow \mathbb{Z}^n \\ \mathbf{x} &\mapsto \sum_{j=0}^n w_{ji} x_j = [W^\top \mathbf{x}]_i \end{aligned}$$

where $[W^\top \mathbf{x}]_i$ represents the i -th row of the product $W^\top \mathbf{x}$.

Definition 1.5 (State). A state E represents a vertex in a transition graph G . It is defined by a series of n Boolean variables, corresponding to the dimension of G . The state E can be represented as:

$$\begin{aligned} E &= x_1 x_2 \dots x_n = (x_i)_{i \in \llbracket 1, n \rrbracket} \quad (\text{binary representation}) \\ &= \sum_{i=1}^n x_i 2^{n-i} \in \llbracket 0, 2^n - 1 \rrbracket \quad (\text{decimal representation}) \end{aligned}$$

Note that $\mathbf{x}^E = (x_1^E, \dots, x_n^E) \in \mathcal{B}^n$ the Boolean variable vector associates with state E .

Definition 1.6 (Successive state). The successive state E_s of a state E is defined by the following equations:

$$\begin{aligned} \mathbf{x}_s^E &= \text{sign} \circ f(\mathbf{x}^E) \\ &= \text{sign} \left(\begin{pmatrix} f_1(\mathbf{x}^E) \\ \vdots \\ f_n(\mathbf{x}^E) \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} [W^\top \mathbf{x}^E]_1 \\ \vdots \\ [W^\top \mathbf{x}^E]_n \end{pmatrix} \right) \\ &= \text{sign}(W^\top \mathbf{x}^E) \end{aligned}$$

Here, $\mathbf{x}_s^E \in \mathcal{B}^n$ is the Boolean variable vector associate with the successive state E_s .

Remark 1.6.1. $\text{sign} \circ f_i(\mathbf{x}^E)$ determines the value of the i -th bit of the state E .

Definition 1.7 (Control bit). A control bit $x_i^E \in \mathbf{x}^E, i \in \llbracket 1, n \rrbracket$ is instrumental in determining the specific sub transition graph G^k to which a state E belongs. Note that the values of the control bits $(x_1^E, \dots, x_m^E)$ are $(k_1, \dots, k_m)$, we define $k = g(k_1, \dots, k_m)$, where g a function depends on m variables.

Example 1.1.

- *one control bit:* $k = g(k_1) = k_1, k \in \mathcal{B}$
- *two control bits:* $k = g(k_1, k_2) = k_1 \oplus k_2, \forall (k_1, k_2) \in \mathcal{B}^2$, where $\oplus$ denotes the XOR

logical operation.

Definition 1.8 (Sub transition graph G^k (one control bit)). A sub transition graph G^k of a transition graph G containing all states E whose i -th bit equal k , $i \in \llbracket 1, n \rrbracket$, which means:

$$\forall E \in G^k \cap G, E = x_1 x_2 \dots \underbrace{k}_{x_i} \dots x_n.$$

Definition 1.9 (Sub transition graph G^k (two control bits)). A sub transition graph G^k of a transition graph G containing all states E whose i -th bit equal k_1 and j -th bit equal k_2 , $(i, j) \in \llbracket 1, n \rrbracket^2, i < j$, which means:

$$\forall E \in G^k \cap G, E = x_1 x_2 \dots \underbrace{k_1}_{x_i} \dots \underbrace{k_2}_{x_j} \dots x_n.$$

Definition 1.10 (Sub transition graph closed). A sub transition graph G^k of a transition graph G is considered closed if, for every state E within G^k , the successive state E_s also belongs to G^k . This can be formally expressed as:

$$\forall E \in G^k, E_s \in G^k.$$

Example 1.2 (Sub transition graph closed). In the following Figure 1.1, G and G^1 is closed since $\forall E \in G^1, E_s \in G^1$, same for G .

Remark 1.10.1. We consider the transition graph G composed of G^k for $k \in \{0, 1\}$ is always closed.

Example 1.3 (Sub transition graph not closed). In the following Figure 1.1, G^0 is not closed since $E = 000 \in G^0, E_s(000) = 001 \in G^1$.

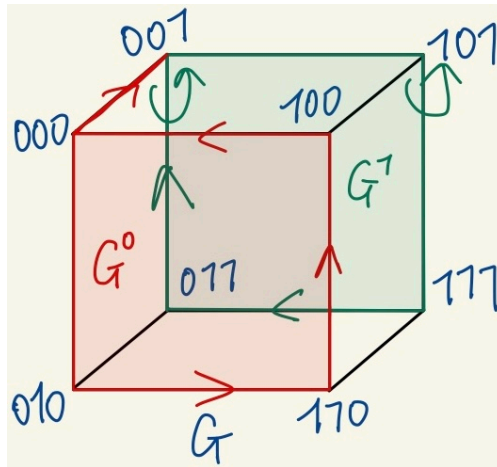


Figure 1.1

Chapter 2

Formulation of the problem

Definition 2.1 (Predicate). For a triplet (G^0, G^1, G) , it satisfies the predicate if and only if the following conditions are met:

- $\dim(G^0) = \dim(G^1) = \dim(G) - 1$
- G^0, G^1 are disjoint (i.e. $G^0 \cap G^1 = \emptyset$).
- G^0, G^1 are closed (cf. Definition 1.8 and 1.10).
- the pluripotent networks (R^0, R^1, R) associated with (G^0, G^1, G) are SBNs.
- G is composed of G^0 and G^1 .

Remark 2.1.1. In the general problem, the dimensions of the sub graphs G^k any dimension less than the dimension of G . To simplify the discussion, we restrict the dimensions of sub graphs G^0 and G^1 to $\dim(G) - 1$. However, if the one of the sub graphs can be contiguously decomposed to a lower dimension, the problem can be extended to a $(m + 1)$ -tuple $(G^0, G^1, \dots, G^m, G)$, with $\dim(G^i) < n, \forall i \in \llbracket 1, m \rrbracket$.

Chapter 3

An efficient representation for Successive State

In this section, we introduce an efficient method to determine the successive state, denoted as E_s , of a given state E in a transition graph G . This method will be employed in subsequent sections. We will start with a 2D application, proceed to 3D scenarios, and ultimately generalize the approach to n-dimensions.

3.1 Computation of E_s in 2D

Given a weight matrix $W \in \mathcal{M}^{2 \times 2}$, a Boolean variable vector $\mathbf{x}^E \in \mathcal{B}^2$ defined as follows:

$$W = \begin{pmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{pmatrix} = \begin{pmatrix} L_1 \\ L_2 \end{pmatrix} \quad \mathbf{x}^E = \begin{pmatrix} x_1^E \\ x_2^E \end{pmatrix}$$

By Definition 1.6, the Boolean variable vector $\mathbf{x}_s^E \in \mathcal{B}^2$, associated with the successive state E_s , is calculated as follows:

$$\begin{aligned} \mathbf{x}_s^E &= \text{sign} \circ f(\mathbf{x}^E) \\ &= \text{sign} \left(\begin{pmatrix} f_1(\mathbf{x}^E) \\ f_2(\mathbf{x}^E) \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11}x_1^E + w_{21}x_2^E \\ w_{12}x_1^E + w_{22}x_2^E \end{pmatrix} \right) \end{aligned}$$

Iterating through all possible states, we derive:

- $E = 00$, $x_1 = x_2 = 0$: the result is always $\begin{pmatrix} 0 & 0 \end{pmatrix}^\top$ for a SBN.
- $E = 10$, $x_1 = 1$, $x_2 = 0$:

$$\mathbf{x}_s^1 = \text{sign} \circ f \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 1 + w_{21} \cdot 0 \\ w_{12} \cdot 1 + w_{22} \cdot 0 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \\ w_{12} \end{pmatrix} \right) = \text{sign}(L_1^\top).$$

- $E = 01, x_1 = 0, x_2 = 1$:

$$\mathbf{x}_s^2 = \text{sign} \circ f \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 0 + w_{21} \cdot 1 \\ w_{12} \cdot 0 + w_{22} \cdot 1 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{21} \\ w_{22} \end{pmatrix} \right) = \text{sign}(L_2^\top).$$

- $E = 11, x_1 = x_2 = 1$:

$$\begin{aligned} \mathbf{x}_s^3 &= \text{sign} \circ f \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 1 + w_{21} \cdot 1 \\ w_{12} \cdot 1 + w_{22} \cdot 1 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11} + w_{21} \\ w_{12} + w_{22} \end{pmatrix} \right) = \text{sign}((L_1 + L_2)^\top). \end{aligned}$$

3.2 Computation of E_s in 3D

Similarly to the 2D case, consider a weight matrix $W \in \mathcal{M}^{3 \times 3}$ and a variable $\mathbf{x}^E \in \mathcal{B}^3$, defined as follows:

$$W = \begin{pmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{pmatrix} = \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} \quad \mathbf{x}^E = \begin{pmatrix} x_1^E \\ x_2^E \\ x_3^E \end{pmatrix}$$

By Definition 1.6, the Boolean variable $\mathbf{x}_s^E \in \mathcal{B}^3$, associated with the successive state E_s , is calculated as follows:

$$\begin{aligned} \mathbf{x}_s^E &= \text{sign} \circ f(\mathbf{x}^E) \\ &= \text{sign} \left(\begin{pmatrix} f_1(\mathbf{x}^E) \\ f_2(\mathbf{x}^E) \\ f_3(\mathbf{x}^E) \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11}x_1^E + w_{21}x_2^E + w_{31}x_3^E \\ w_{12}x_1^E + w_{22}x_2^E + w_{32}x_3^E \\ w_{13}x_1^E + w_{23}x_2^E + w_{33}x_3^E \end{pmatrix} \right) \end{aligned}$$

Iterating through all possible states, we derive:

- $E = 000$: the result is always $(0 \ 0 \ 0)^\top$ for a SBN.
- $E = 100$:

$$\begin{aligned} \mathbf{x}_s^1 &= \text{sign} \circ f \left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 1 + w_{21} \cdot 0 + w_{31} \cdot 0 \\ w_{12} \cdot 1 + w_{22} \cdot 0 + w_{32} \cdot 0 \\ w_{13} \cdot 1 + w_{23} \cdot 0 + w_{33} \cdot 0 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11} \\ w_{12} \\ w_{13} \end{pmatrix} \right) = \text{sign}(L_1^\top) \end{aligned}$$

- $E = 010$:

$$\begin{aligned}\mathbf{x}_s^2 &= \text{sign} \circ f \left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 0 + w_{21} \cdot 1 + w_{31} \cdot 0 \\ w_{12} \cdot 0 + w_{22} \cdot 1 + w_{32} \cdot 0 \\ w_{13} \cdot 0 + w_{23} \cdot 1 + w_{33} \cdot 0 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{12} \\ w_{22} \\ w_{32} \end{pmatrix} \right) = \text{sign}(L_2^\top)\end{aligned}$$

- $E = 110$:

$$\begin{aligned}\mathbf{x}_s^3 &= \text{sign} \circ f \left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 1 + w_{21} \cdot 1 + w_{31} \cdot 0 \\ w_{12} \cdot 1 + w_{22} \cdot 1 + w_{32} \cdot 0 \\ w_{13} \cdot 1 + w_{23} \cdot 1 + w_{33} \cdot 0 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11} + w_{12} \\ w_{12} + w_{22} \\ w_{13} + w_{23} \end{pmatrix} \right) = \text{sign}((L_1 + L_2)^\top)\end{aligned}$$

- $E = 001$:

$$\begin{aligned}\mathbf{x}_s^4 &= \text{sign} \circ f \left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 0 + w_{21} \cdot 0 + w_{31} \cdot 1 \\ w_{12} \cdot 0 + w_{22} \cdot 0 + w_{32} \cdot 1 \\ w_{13} \cdot 0 + w_{23} \cdot 0 + w_{33} \cdot 1 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{13} \\ w_{23} \\ w_{33} \end{pmatrix} \right) = \text{sign}(L_3^\top)\end{aligned}$$

- $E = 101$:

$$\begin{aligned}\mathbf{x}_s^5 &= \text{sign} \circ f \left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 1 + w_{21} \cdot 0 + w_{31} \cdot 1 \\ w_{12} \cdot 1 + w_{22} \cdot 0 + w_{32} \cdot 1 \\ w_{13} \cdot 1 + w_{23} \cdot 0 + w_{33} \cdot 1 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11} + w_{13} \\ w_{21} + w_{23} \\ w_{31} + w_{33} \end{pmatrix} \right) = \text{sign}((L_1 + L_3)^\top)\end{aligned}$$

- $E = 011$:

$$\begin{aligned}\mathbf{x}_s^6 &= \text{sign} \circ f \left(\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 0 + w_{21} \cdot 1 + w_{31} \cdot 1 \\ w_{12} \cdot 0 + w_{22} \cdot 1 + w_{32} \cdot 1 \\ w_{13} \cdot 0 + w_{23} \cdot 1 + w_{33} \cdot 1 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{12} + w_{13} \\ w_{22} + w_{23} \\ w_{32} + w_{33} \end{pmatrix} \right) = \text{sign}((L_2 + L_3)^\top)\end{aligned}$$

- $E = 111$:

$$\begin{aligned}\mathbf{x}_s^7 &= \text{sign} \circ f \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11} \cdot 1 + w_{21} \cdot 1 + w_{31} \cdot 1 \\ w_{12} \cdot 1 + w_{22} \cdot 1 + w_{32} \cdot 1 \\ w_{13} \cdot 1 + w_{23} \cdot 1 + w_{33} \cdot 1 \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11} + w_{12} + w_{13} \\ w_{21} + w_{22} + w_{23} \\ w_{31} + w_{32} + w_{33} \end{pmatrix} \right) = \text{sign}((L_1 + L_2 + L_3)^\top)\end{aligned}$$

3.3 Generalization to n-Dimensions

Proposition 3.1. *Given $\mathbf{x}^E \in \mathcal{B}^n$, G a transition graph, $W \in \mathcal{M}^{n \times n}$ the weight matrix associate with G , $L_i = (w_{i1}, \dots, w_{in})$ the i -th row of W , $i \in \llbracket 1, n \rrbracket$, then*

$$\forall E \in G, \mathbf{x}_s^E = \text{sign} \circ f(\mathbf{x}^E) = \text{sign} \left(\left(\sum_{i=1}^n L_i x_i^E \right)^\top \right).$$

Proof. Similarly to the 2D and 3D case, consider a weight matrix $W \in \mathcal{M}^{n \times n}$ and a variable $\mathbf{x} \in \mathcal{B}^n$, defined as follows:

$$W = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & w_{nn} \end{pmatrix} = \begin{pmatrix} L_1 \\ \vdots \\ L_n \end{pmatrix} \quad \mathbf{x}^E = \begin{pmatrix} x_1^E \\ \vdots \\ x_n^E \end{pmatrix}$$

By Definition 1.6, the Boolean variable $\mathbf{x}_s^E \in \mathcal{B}^n$, associated with the successive state E_s , is

calculated as follows:

$$\begin{aligned}
\mathbf{x}_s^E &= \text{sign} \circ f(\mathbf{x}^E) \\
&= \text{sign} \left(\begin{pmatrix} f_1(\mathbf{x}^E) \\ \vdots \\ f_n(\mathbf{x}^E) \end{pmatrix} \right) \\
&= \text{sign} \left(\begin{pmatrix} \sum_{i=1}^n w_{1i} x_i^E \\ \vdots \\ \sum_{i=1}^n w_{ni} x_i^E \end{pmatrix} \right) \\
&= \text{sign} \left(\sum_{i=1}^n \begin{pmatrix} w_{1i} \\ \vdots \\ w_{ni} \end{pmatrix} \cdot x_i^E \right) \\
&= \text{sign} \left(\sum_{i=1}^n L_i^\top x_i^E \right) \\
&= \text{sign} \left(\left(\sum_{i=1}^n L_i x_i^E \right)^\top \right) \quad \text{since } x_i^E \text{ is a constant}
\end{aligned}$$

■

Part II

Constraints, Properties of weight
matrices and Composition method (1
control bit)

Chapter 4

Closure Constraint

For every triplet (G^0, G^1, G) that satisfies the predicate, G^0 and G^1 are closed. The Closure Constraints can be derived from Definition 1.10. Same as the previous section, we will start with a 2D application, proceed to 3D scenarios, and ultimately generalize the approach to n-dimensions.

In the subsequent content, we will use x_n as the control bit to simplify representations. However, the methodology is applicable to any control bit x_i , $i \in \llbracket 1, n \rrbracket$.

4.1 Closure constraints 2D

Recall the definition of closure:

$$\forall k \in \llbracket 0, 1 \rrbracket, \forall E \in G^k, E_s \in G^k.$$

Given that the weight matrix W and the Boolean variable $\mathbf{x}^E$ are defined as follows:

$$W = \begin{pmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{pmatrix} \quad \mathbf{x}^E = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

By Definition 1.6, the Boolean variable $\mathbf{x}_s^E \in \mathcal{B}^2$, associated with the successive state E_s , is calculated as follows:

$$\begin{aligned} \mathbf{x}_s^E &= \text{sign} \circ f(\mathbf{x}^E) \\ &= \text{sign} \left(\begin{pmatrix} f_1(\mathbf{x}^E) \\ f_2(\mathbf{x}^E) \end{pmatrix} \right) = \text{sign} \left(\begin{pmatrix} w_{11}x_1 + w_{21}x_2 \\ w_{12}x_1 + w_{22}x_2 \end{pmatrix} \right) \end{aligned}$$

In 2D, the definition of closure implies that the second bit of each state E consistently equals k :

$$\forall x_1 \in \mathcal{B}, \text{sign} \circ f_2 \left(\begin{pmatrix} x_1 \\ k \end{pmatrix} \right) = k \iff \begin{cases} \text{sign} \circ f_2 \left(\begin{pmatrix} x_1 \\ 0 \end{pmatrix} \right) = 0 \\ \text{sign} \circ f_2 \left(\begin{pmatrix} x_1 \\ 1 \end{pmatrix} \right) = 1 \end{cases}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for both $k = 0$ and $k = 1$. Consequently, we examine their intersection:

$$\begin{aligned} \forall x_1 \in \mathcal{B}, & \left(\text{sign} \circ f_2 \left(\begin{pmatrix} x_1 \\ 0 \end{pmatrix} \right) = 0 \right) \wedge \left(\text{sign} \circ f_2 \left(\begin{pmatrix} x_1 \\ 1 \end{pmatrix} \right) = 1 \right) \\ & \iff (w_{12}x_1 + w_{22} \cdot 0 \leq 0) \wedge (w_{12}x_1 + w_{22} \cdot 1 > 0) \\ & \iff (w_{12}x_1 \leq 0) \wedge (w_{12}x_1 + w_{22} > 0) \end{aligned}$$

Enumerating all possible values of x_1 to more precisely define the constraints:

- $x_1 = 0$:

$$(w_{11} \cdot 0 \leq 0) \wedge (w_{22} > 0) \iff w_{22} > 0 \quad (4.1)$$

- $x_1 = 1$:

$$(w_{12} \cdot 1 \leq 0) \wedge (w_{12} \cdot 1 + w_{22} \cdot 1 > 0) \iff (w_{12} \leq 0) \wedge (w_{22} > |w_{12}| \geq 0) \quad (4.2)$$

By combining Constraint (4.1) and Constraint (4.2), we can derive the closure constraints for 2D:

$$(w_{12} \leq 0) \wedge (w_{22} > -w_{12}) \iff (w_{12} \leq 0) \wedge (w_{22} > |w_{12}|).$$

4.2 Closure constraints 3D

Similar as 2D, given that the weight matrix W and the Boolean variable $\mathbf{x}^E$ are defined as follows:

$$W = \begin{pmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{pmatrix} \quad \mathbf{x}^E = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

By Definition 1.6, the Boolean variable $\mathbf{x}_s^E \in \mathcal{B}^3$, associated with the successive state E_s , is calculated as follows:

$$\begin{aligned} \mathbf{x}_s^E &= \text{sign} \circ f(\mathbf{x}^E) \\ &= \text{sign} \left(\begin{pmatrix} f_1(\mathbf{x}^E) \\ f_2(\mathbf{x}^E) \\ f_3(\mathbf{x}^E) \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} w_{11}x_1 + w_{21}x_2 + w_{31}x_3 \\ w_{12}x_1 + w_{22}x_2 + w_{32}x_3 \\ w_{13}x_1 + w_{23}x_2 + w_{33}x_3 \end{pmatrix} \right) \end{aligned}$$

In 3D, the definition of closure implies that the third bit of each state E consistently equals the control bit k :

$$\forall (x_1, x_2) \in \mathcal{B}^2, \text{sign} \circ f_3 \left(\begin{pmatrix} x_1 \\ x_2 \\ k \end{pmatrix} \right) = k \iff \begin{cases} \text{sign} \circ f_3 \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} \right) = 0 \\ \text{sign} \circ f_3 \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix} \right) = 1 \end{cases}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for both $k = 0$ and $k = 1$. Consequently, we examine their intersection:

$$\begin{aligned} & (w_{13}x_1 + w_{23}x_2 + w_{33} \cdot 1 > 0) \wedge (w_{13}x_1 + w_{23}x_2 + w_{33} \cdot 0 \leq 0) \\ \iff & (w_{13}x_1 + w_{23}x_2 + w_{33} > 0) \wedge (w_{13}x_1 + w_{23}x_2 \leq 0) \end{aligned}$$

Enumerating all possible values of tuple (x_1, x_2) to more precisely define the constraints:

- $x_1 = 0, x_2 = 0$:

$$(w_{13} \cdot 0 + w_{23} \cdot 0 \leq 0) \wedge (w_{33} > 0) \iff w_{33} > 0 \quad (4.3)$$

- $x_1 = 0, x_2 = 1$:

$$\begin{aligned} & (w_{13} \cdot 0 + w_{23} \cdot 1 \leq 0) \wedge (w_{13} \cdot 0 + w_{23} \cdot 1 + w_{33} > 0) \\ \iff & (w_{23} \leq 0) \wedge (w_{33} > |w_{23}|) \end{aligned} \quad (4.4)$$

- $x_1 = 1, x_2 = 0$:

$$\begin{aligned} & (w_{13} \cdot 1 + w_{23} \cdot 0 \leq 0) \wedge (w_{13} \cdot 1 + w_{23} \cdot 0 + w_{33} > 0) \\ \iff & (w_{13} \leq 0) \wedge (w_{33} > |w_{13}|) \end{aligned} \quad (4.5)$$

- $x_1 = 1, x_2 = 1$:

$$\begin{aligned} & (w_{13} \cdot 1 + w_{23} \cdot 1 \leq 0) \wedge (w_{13} \cdot 1 + w_{23} \cdot 1 + w_{33} > 0) \\ \iff & 1 \wedge (w_{33} > -(w_{13} + w_{23})) \text{ since } w_{13} \leq 0, w_{23} \leq 0 \\ \iff & w_{33} > |w_{13} + w_{23}| = |w_{13}| + |w_{23}| \end{aligned} \quad (4.6)$$

By combining Constraint (4.3), (4.4), (4.5) and (4.6) we can derive the closure constraints for 3D:

$$(w_{13} \leq 0) \wedge (w_{23} \leq 0) \wedge (w_{33} > |w_{13}| + |w_{23}|)$$

Example 4.1. Let $E = 100 \in G^0$, we have

$$W = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \mathbf{x}^1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{x}_s^1 = \text{sign} \circ f(\mathbf{x}^1) = \text{sign} \left(\begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Therefore, $E_s = 100 = E \in G^0$, satisfying the closure constraints.

Example 4.2 (Counter-example). Keeping the same state E whiling utilizing a different weight matrix

$$W' = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

The matrix W' does not satisfy the Closure Constraints because $w_{13} = 1 > 0$, $w_{23} = 1 > 0$ and $w_{33} < |w_{13}| + |w_{23}|$. By the definition

$$\mathbf{x}_s^1 = \text{sign} \circ f(\mathbf{x}^1) = \text{sign} \left(\begin{pmatrix} 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 0 + 1 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$E_s = 101 \in G^1$, which does not adhere to the closure constraints.

4.3 Generalized Closure Constraints

Proposition 4.1 (Generalized closure constraints). For a triplet (G^0, G^1, G) that satisfies the predicate, the following closure constraints must be met:

$$\begin{aligned} & \left(\bigwedge_{i=1}^{n-1} w_{in} \leq 0 \right) \wedge \left(w_{nn} > \sum_{i=1}^{n-1} |w_{in}| \right) \\ \iff & (w_{1n} \leq 0) \wedge \cdots \wedge (w_{(n-1)n} \leq 0) \wedge \left(w_{nn} > \sum_{i=1}^{n-1} |w_{in}| \right) \end{aligned}$$

Proof. Similar as 2D and 3D, given that the weight matrix W and the Boolean variable $\mathbf{x}^E \in \mathcal{B}^n$ are defined as follows:

$$W = \begin{pmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & w_{nn} \end{pmatrix} \quad \mathbf{x}^E = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

By Definition 1.6, the Boolean variable $\mathbf{x}_s^E \in \mathcal{B}^3$, associated with the successive state E_s , is calculated as follows:

$$\begin{aligned}\mathbf{x}_s^E &= \text{sign} \circ f(\mathbf{x}^E) \\ &= \text{sign} \left(\begin{pmatrix} f_1(\mathbf{x}^E) \\ \vdots \\ f_n(\mathbf{x}^E) \end{pmatrix} \right) \\ &= \text{sign} \left(\begin{pmatrix} \sum_{i=1}^n w_{1i}x_i \\ \vdots \\ \sum_{i=1}^n w_{ni}x_i \end{pmatrix} \right)\end{aligned}$$

In n-Dimension, the definition of closure implies that the n-th bit of each state E consistently equals k :

$$\forall (x_1, \dots, x_{n-1}) \in \mathcal{B}^{n-1}, \text{sign} \circ f_n \left(\begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \\ k \end{pmatrix} \right) = k \iff \begin{cases} \text{sign} \circ f_n \left(\begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \\ 0 \end{pmatrix} \right) = 0 \\ \text{sign} \circ f_n \left(\begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \\ 1 \end{pmatrix} \right) = 1 \end{cases}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for both $k = 0$ and $k = 1$. Consequently, we examine their intersection:

$$\begin{aligned}& \left(\sum_{i=1}^{n-1} w_{in}x_i + w_{nn} \cdot 0 \leq 0 \right) \wedge \left(\sum_{i=1}^{n-1} w_{in}x_i + w_{nn} \cdot 1 > 0 \right) \\ & \iff \left(\sum_{i=1}^{n-1} w_{in}x_i \leq 0 \right) \wedge \left(\sum_{i=1}^{n-1} w_{in}x_i + w_{nn} > 0 \right)\end{aligned}$$

To more precisely define the constraints, note that S the set of x_i which value is 1 and $\#S$ the cardinal of the set S , then:

- $\#S = 0, x_1 = \dots = x_{n-1} = 0$:

$$(0 \leq 0) \wedge (w_{nn} > 0) \iff w_{nn} > 0 \quad (4.7)$$

- $\#S = 1, (i, j) \in \llbracket 1, n-1 \rrbracket^2, \forall j \neq i, x_i = 1, x_j = 0$:

$$\begin{aligned} & (w_{in} \cdot 1 \leq 0) \wedge (w_{in} \cdot 1 + w_{nn} > 0) \\ \iff & (w_{in} \leq 0) \wedge (w_{nn} > |w_{in}|) \end{aligned} \quad (4.8)$$

- $\#S = 2, \dots, \#S = n-1$: Discussion of these cases is unnecessary as they can be inferred from the scenarios $\#S = 1$ and $\#S = n$.
- $\#S = n, x_1 = \dots = x_{n-1} = 1$:

$$\begin{aligned} & \left(\sum_{i=1}^{n-1} w_{in} \cdot 1 \leq 0 \right) \wedge \left(\sum_{i=1}^{n-1} w_{in} \cdot 1 + w_{nn} > 0 \right) \\ \iff & \left(\sum_{i=1}^{n-1} w_{in} \leq 0 \right) \wedge \left(w_{nn} > \left| \sum_{i=1}^{n-1} w_{in} \right| \right) \\ \iff & \left(\sum_{i=1}^{n-1} w_{in} \leq 0 \right) \wedge \left(w_{nn} > \sum_{i=1}^{n-1} |w_{in}| \right) \end{aligned} \quad (4.9)$$

Since $\forall i \in \llbracket 1, n-1 \rrbracket, w_{in} \leq 0$ by Constraint (4.8).

By combining Constraint (4.7), (4.8) and (4.9) we can derive the closure constraints for n dimensions:

$$\begin{aligned} & (w_{1n} \leq 0) \wedge \dots \wedge (w_{(n-1)n} \leq 0) \wedge \left(w_{nn} > \sum_{i=1}^{n-1} |w_{in}| \right) \\ \iff & \bigwedge_{i=1}^{n-1} (w_{in} \leq 0) \wedge \left(w_{nn} > \sum_{i=1}^{n-1} |w_{in}| \right) \end{aligned}$$

■

Chapter 5

SBN Constraints

Let (G^0, G^1, G) be a triplet verifies the predicate defined in Definition 2.1, the following SBN Constraints apply:

- R^0, R^1, R associated to G^0, G^1, G are SBNs.
- G^0 and G^1 exhibit the same behavior as G , since G is composed of them.

Note that the associated weight matrices W^0, W^1, W , and the Boolean functions f^0, f^1, f for G^0, G^1 and G respectively.

Same as the previous chapter, we fix x_n as the control bit to simplify representations. The methodology is applicable to any control bit $x_i, i \in \llbracket 1, n \rrbracket$.

5.1 SBN Constraints 2D

In 2D, $\forall k \in \{0, 1\}, G^k$ associates with R^k is SBN, and G^k has the same behaviors as G gives

$$\begin{aligned} \forall x_1 \in \mathcal{B}, \mathbf{x}_s^E &= \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ k \end{pmatrix} \right) = \text{sign} \circ f_1^k(x_1) \\ \iff &\begin{cases} \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ 0 \end{pmatrix} \right) = \text{sign} \circ f_1^0(x_1) \\ \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ 1 \end{pmatrix} \right) = \text{sign} \circ f_1^1(x_1) \end{cases} \end{aligned}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for both $k = 0$ and $k = 1$. Consequently, we examine their intersection:

$$\begin{aligned} &(\text{sign}(w_{11}x_1 + w_{21} \cdot 0) = \text{sign}(w_{11}^0x_1)) \wedge (\text{sign}(w_{11}x_1 + w_{21} \cdot 1) = \text{sign}(w_{11}^1x_1)) \\ \iff &(\text{sign}(w_{11}x_1) = \text{sign}(w_{11}^0x_1)) \wedge (\text{sign}(w_{11}x_1 + w_{21}) = \text{sign}(w_{11}^1x_1)) \end{aligned}$$

Enumerating all possible values of tuple x_1 to more precisely define the constraints:

- $x_1 = 0$:

$$(sign(0) = sign(0)) \wedge (sign(w_{21}) = sign(0)) \iff w_{21} \leq 0 \quad (5.1)$$

- $x_1 = 1$:

$$(sign(w_{11}) = sign(w_{11}^0)) \wedge (sign(w_{11} + w_{21}) = sign(w_{11}^1)) \quad (5.2)$$

By combining Constraint (5.1) and Constraint (5.2), we can derive the SBN constraints for 2D:

$$(w_{21} \leq 0) \wedge (sign(w_{11}) = sign(w_{11}^0)) \wedge (sign(w_{11} + w_{21}) = sign(w_{11}^1))$$

5.2 SBN Constraints 3D

In 3D, $\forall k \in \{0, 1\}$, G^k associates with R^k is SBN, and G^k has the same behaviors as G gives

$$\begin{aligned} \forall (x_1, x_2) \in \mathcal{B}^2, \mathbf{x}_s^E = sign \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ k \end{pmatrix} \right) &= sign \circ f_i^k \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) \\ \iff \begin{cases} sign \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} \right) &= sign \circ f_i^0 \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) \\ sign \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \end{pmatrix} \right) &= sign \circ f_i^1 \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) \end{cases} \end{aligned}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for both $k = 0$ and $k = 1$. Consequently, we examine their intersection:

$$\begin{aligned} \forall i \in \llbracket 1, 2 \rrbracket, (sign(w_{1i}x_1 + w_{2i}x_2 + w_{3i} \cdot 0) &= sign(w_{1i}^0x_1 + w_{2i}^0x_2)) \\ \wedge (sign(w_{1i}x_1 + w_{2i}x_2 + w_{3i} \cdot 1) &= sign(w_{1i}^1x_1 + w_{2i}^1x_2)) \\ \iff (sign(w_{1i}x_1 + w_{2i}x_2) &= sign(w_{1i}^0x_1 + w_{2i}^0x_2)) \\ \wedge (sign(w_{1i}x_1 + w_{2i}x_2 + w_{3i}) &= sign(w_{1i}^1x_1 + w_{2i}^1x_2)) \end{aligned}$$

Enumerating all possible values of tuple $(x_1, x_2) \in \mathcal{B}^2$ to more precisely define the constraints:

- $x_1 = x_2 = 0$:

$$\begin{aligned} (sign(w_{1i} \cdot 0 + w_{2i} \cdot 0) &= sign(w_{1i}^0 \cdot 0 + w_{2i}^0 \cdot 0)) \\ \wedge (sign(w_{1i} \cdot 0 + w_{2i} \cdot 0 + w_{3i}) &= sign(w_{1i}^1 \cdot 0 + w_{2i}^1 \cdot 0)) \\ \iff (sign(0) = sign(0)) \wedge (sign(w_{3i}) &= sign(0)) \\ \iff w_{3i} \leq 0 \end{aligned} \quad (5.3)$$

- $x_1 = 1, x_2 = 0$:

$$\begin{aligned}
& (\text{sign}(w_{1i} \cdot 1 + w_{2i} \cdot 0) = \text{sign}(w_{1i}^0 \cdot 1 + w_{2i}^0 \cdot 0)) \\
& \wedge (\text{sign}(w_{1i} \cdot 1 + w_{2i} \cdot 0 + w_{3i}) = \text{sign}(w_{1i}^1 \cdot 1 + w_{2i}^1 \cdot 0)) \\
& \iff (\text{sign}(w_{1i}) = \text{sign}(w_{1i}^0)) \wedge (\text{sign}(w_{1i} + w_{3i}) = \text{sign}(w_{1i}^1))
\end{aligned} \tag{5.4}$$

- $x_1 = 0, x_2 = 1$:

$$\begin{aligned}
& (\text{sign}(w_{1i} \cdot 0 + w_{2i} \cdot 1) = \text{sign}(w_{1i}^0 \cdot 0 + w_{2i}^0 \cdot 1)) \\
& \wedge (\text{sign}(w_{1i} \cdot 0 + w_{2i} \cdot 1 + w_{3i}) = \text{sign}(w_{1i}^1 \cdot 0 + w_{2i}^1 \cdot 1)) \\
& \iff (\text{sign}(w_{2i}) = \text{sign}(w_{2i}^0)) \wedge (\text{sign}(w_{2i} + w_{3i}) = \text{sign}(w_{2i}^1))
\end{aligned} \tag{5.5}$$

- $x_1 = x_2 = 1$:

$$\begin{aligned}
& (\text{sign}(w_{1i} \cdot 1 + w_{2i} \cdot 1) = \text{sign}(w_{1i}^0 \cdot 1 + w_{2i}^0 \cdot 1)) \\
& \wedge (\text{sign}(w_{1i} \cdot 1 + w_{2i} \cdot 1 + w_{3i}) = \text{sign}(w_{1i}^1 \cdot 1 + w_{2i}^1 \cdot 1)) \\
& \iff (\text{sign}(w_{1i} + w_{2i}) = \text{sign}(w_{1i}^0 + w_{2i}^0)) \\
& \wedge (\text{sign}(w_{1i} + w_{2i} + w_{3i}) = \text{sign}(w_{1i}^1 + w_{2i}^1))
\end{aligned} \tag{5.6}$$

By combining Constraint (5.3), (5.4), (5.5) and (5.6), we can derive the SBN constraints for 3D:

$$\begin{aligned}
& \forall i \in \llbracket 1, 2 \rrbracket, (w_{3i} \leq 0) \\
& \wedge (\text{sign}(w_{1i}) = \text{sign}(w_{1i}^0)) \\
& \wedge (\text{sign}(w_{1i} + w_{3i}) = \text{sign}(w_{1i}^1)) \\
& \wedge (\text{sign}(w_{2i}) = \text{sign}(w_{2i}^0)) \\
& \wedge (\text{sign}(w_{2i} + w_{3i}) = \text{sign}(w_{2i}^1)) \\
& \wedge (\text{sign}(w_{1i} + w_{2i}) = \text{sign}(w_{1i}^0 + w_{2i}^0)) \\
& \wedge (\text{sign}(w_{1i} + w_{2i} + w_{3i}) = \text{sign}(w_{1i}^1 + w_{2i}^1))
\end{aligned}$$

5.3 Generalized SBN Constraints

Proposition 5.1 (Generalized SBN Constraints for 1 control variable). $\forall j \in \llbracket 1, n-1 \rrbracket$ and $\forall \mathbf{x}^E \in \mathcal{B}^n$,

$$\text{sign} \left(\sum_{i=1}^n w_{ij} x_i \right) = \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^{x_n} x_i \right)$$

In particular, $w_{nj} \leq 0$.

Proof. Similarly as in 2D and 3D, G^k have the same behaviors as G gives

$$\forall (x_1, \dots, x_{n-1}) \in \mathcal{B}^{n-1}, \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \\ k \end{pmatrix} \right) = \text{sign} \circ f_i^k \left(\begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \end{pmatrix} \right)$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above $\forall k \in \{0, 1\}$. Consequently, we examine their intersection:

$$\begin{aligned} \text{sign}(w_{1j}x_1 + \dots + w_{(n-1)j}x_{n-1} + w_{nj}k) &= \text{sign}(w_{1j}^kx_1 + \dots + w_{(n-1)j}^kx_{n-1}) \\ \iff \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}x_i + w_{nj}k \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^kx_i \right) \end{aligned}$$

Note that S the set of x_i which value is 1 and $\#S$ the cardinal of the set S , then:

- $\#S = 0 \iff x_1 = \dots = x_{n-1} = 0$:

$$\begin{aligned} \bigwedge_{j=1}^{n-1} \bigwedge_{k=0}^1 \text{sign} \left(\sum_{i=1}^{n-1} w_{ij} \cdot 0 + w_{nj}k \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^k \cdot 0 \right) \\ \bigwedge_{j=1}^{n-1} \bigwedge_{k=0}^1 \text{sign}(w_{nj}k) &= 0 \\ \iff \bigwedge_{j=1}^{n-1} \bigwedge_{k=0}^1 w_{nj}k &\leq 0 \end{aligned} \tag{5.7}$$

- $\#S = 1 \iff x_a = 1, \forall b \neq a, x_b = 0$:

$$\begin{aligned} \bigwedge_{a=1}^{n-1} \bigwedge_{j=1}^{n-1} \bigwedge_{k=0}^1 \text{sign} \left(\sum_{i=1, i \neq a}^{n-1} w_{ij} \cdot 0 + w_{aj} + w_{nj}k \right) &= \text{sign} \left(\sum_{i=1, i \neq a}^{n-1} w_{ij}^k \cdot 0 + w_{aj}^k \right) \\ \bigwedge_{i=1}^{n-1} \bigwedge_{j=1}^{n-1} \bigwedge_{k=0}^1 \text{sign}(w_{aj} + w_{nj}k) &= \text{sign}(w_{aj}^k) \end{aligned} \tag{5.8}$$

- $\#S = 2, \dots, \#S = n - 2$:

$$\vdots \tag{5.9}$$

- $\#S = n - 1 \iff x_1 = \dots = x_{n-1} = 1$:

$$\begin{aligned}
\bigwedge_{j=1}^{n-1} \bigwedge_{k=0}^1 \text{sign} \left(\sum_{i=1}^{n-1} w_{ij} \cdot 1 + w_{nj}k \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^k \cdot 1 \right) \\
\bigwedge_{j=1}^{n-1} \bigwedge_{k=0}^1 \text{sign} \left(\sum_{i=1}^{n-1} w_{ij} + w_{nj}k \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^k \right)
\end{aligned} \tag{5.10}$$

By combining constraint (5.7), (5.8), (5.9) and (5.10), we can derive the SBN constraints for n dimensions:

$$\begin{aligned}
\forall (x_1, \dots, x_{n-1}) \in \mathcal{B}^{n-1}, \bigwedge_{k=0}^1 \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}x_i + w_{nj}k \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^k x_i \right) \\
\iff \forall \mathbf{x}^{\mathbf{E}} \in \mathcal{B}^n, \text{sign} \left(\sum_{i=1}^n w_{ij}x_i \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^{x_n} x_i \right)
\end{aligned}$$

■

Chapter 6

Properties of weight matrices

According to the Proposition (3.1) in Section 3, we can deduce specific properties of the weight matrix W^0 and W^1 .

Recall the generalized form of E_s : for a given transition graph G associated with a weight matrix W , where L_i is the i -th line of W , the Boolean variable vector $\mathbf{x}_s^E \in \mathcal{B}^n$ associated with the successive state E_s is determined by:

$$\mathbf{x}_s^E = \text{sign} \circ f(\mathbf{x}^E) = \text{sign} \left(\left(\sum_{i=1}^n L_i x_i^E \right)^\top \right).$$

Same as the previous chapter, let x_n as the control bit.

6.1 Property of W^0 , W^1 and W

From this basis, we can precisely identify the following two properties:

Property 6.1 (Property of W^0). *Given a triplet (G^0, G^1, G) that satisfies the predicate with associated weight matrices (W^0, W^1, W) , then W^0 exhibits the following characteristics:*

$$W^0 \equiv \begin{pmatrix} w_{11} & \cdots & w_{1(n-1)} \\ \vdots & \ddots & \vdots \\ w_{(n-1)1} & \cdots & w_{(n-1)(n-1)} \end{pmatrix} = W|_{\dim=n-1}.$$

Proof. Let $x_n^E = 0$, yielding $E = x_1 x_2 \dots x_{n-1} 0$. As E belongs to both G^0 and G , we can

apply the Proposition (3.1), then in G we have

$$\begin{aligned}\mathbf{x}_s^E &= \text{sign} \left(\left(\sum_{i=1}^n L_i x_i^E \right)^\top \right) \\ &= \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i x_i^E \right)^\top \right) \quad \text{since } x_n^E = 0\end{aligned}\tag{6.1}$$

Since E is also a member of G^0 , and recognizing that L_i^0 the i -th row of W^0 , then in G^0 we have

$$\mathbf{x}_s^E = \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i^0 x_i^E \right)^\top \right)\tag{6.2}$$

As G^0 is a sub graph of G , G^0 must exhibit the same behaviors as G . This leads us to conclude that Equation (6.1) and Equation (6.2) are equivalent. Then we get

$$\begin{aligned}\text{sign} \left(\left(\sum_{i=1}^{n-1} L_i^0 x_i^E \right)^\top \right) &= \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i x_i^E \right)^\top \right) \\ \iff \sum_{i=1}^{n-1} L_i^0 x_i^E &\equiv \sum_{i=1}^{n-1} L_i x_i^E \quad \forall \mathbf{x}^E \in \mathcal{B}^{n-1}\end{aligned}$$

By iterating over all possible configurations of $\mathbf{x}^E \in \mathcal{B}^n$, we establish that $W^0 \equiv W|_{\dim=n-1}$. ■

Property 6.2 (Property of W^1). *Given a triplet (G^0, G^1, G) that satisfies the predicate with associated weight matrices (W^0, W^1, W) , then W^1 exhibits the following characteristics:*

$$W^1 \equiv \left(\begin{array}{c} L_1 + L_n \\ \vdots \\ L_{n-1} + L_n \end{array} \right) \Big|_{\dim=n-1}.$$

Proof. Let $x_n^E = 1$, yielding $E = x_1 x_2 \dots x_{n-1} 1$. As E belongs to both G^1 and G , we can apply the Proposition (3.1), then in G we have

$$\begin{aligned}\mathbf{x}_s^E &= \text{sign} \left(\left(\sum_{i=1}^n L_i x_i^E \right)^\top \right) \\ &= \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i x_i^E + L_n \right)^\top \right) \quad \text{since } x_n^E = 1\end{aligned}\tag{6.3}$$

Since E is also a member of G^1 , and recognizing that L_i^1 the i -th row of W^1 , then in G^1 we have

$$\mathbf{x}_s^E = \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i^1 x_i^E \right)^\top \right) \quad (6.4)$$

As G^1 is a sub graph of G , G^1 must exhibit the same behaviors as G . This leads us to conclude that Equation (6.3) and Equation (6.4) are equivalent. Then we get

$$\begin{aligned} \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i^1 x_i^E \right)^\top \right) &= \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i x_i^E + L_n \right)^\top \right) \\ \iff \sum_{i=1}^{n-1} L_i^1 x_i^E &\equiv \sum_{i=1}^{n-1} L_i x_i^E + L_n \quad \forall i \in \llbracket 1, n-1 \rrbracket, x_i^E \in \mathcal{B} \end{aligned}$$

By iterating over all possible configurations of $\mathbf{x}^E \in \mathcal{B}^n$, we establish that

$$W^1 \equiv \left(\begin{array}{c} L_1 + L_n \\ \vdots \\ L_{n-1} + L_n \end{array} \right) \Big|_{\dim=n-1}.$$

■

Proposition 6.1 (Structure of matrix W for 1 control bit). *We can construct the weight matrix W for G applying Property 6.1 and 6.2 alongside the closure constraints discussed in Section 4 and the SBN constraints outlined in Section 5.*

$$W = \left(\begin{array}{ccc|c} \boxed{w_{11} \quad \cdots \quad w_{1(n-1)}} & \boxed{w_{1n} \leq 0} \\ \vdots & \vdots \\ \boxed{w_{(n-1)1} \quad \cdots \quad w_{(n-1)(n-1)}} & \boxed{w_{(n-1)n} \leq 0} \\ \boxed{w_{n1} \leq 0 \quad \cdots \quad w_{n(n-1)} \leq 0} & \boxed{w_{nn} > \sum_{i=1}^{n-1} |w_{in}|} \end{array} \right) = \left(\begin{array}{c} L_1 \\ \vdots \\ L_{n-1} \\ L_n \end{array} \right)$$

where

$$W^0 \equiv \left(\begin{array}{ccc} \boxed{w_{11} \quad \cdots \quad w_{1(n-1)}} \\ \vdots \\ \boxed{w_{(n-1)1} \quad \cdots \quad w_{(n-1)(n-1)}} \end{array} \right) \quad W^1 \equiv \left(\begin{array}{c} L_1 + L_n \\ \vdots \\ L_{n-1} + L_n \end{array} \right) \Big|_{\dim=n-1}$$

In particular, we can directly take $w_{in} = 0 \quad \forall i \in \llbracket 1, n-1 \rrbracket$ and $w_{nn} = 1$ to simplify the

computation.

$$W = \begin{pmatrix} \begin{matrix} w_{11} & \cdots & w_{1(n-1)} \\ \vdots & \ddots & \vdots \\ w_{(n-1)1} & \cdots & w_{(n-1)(n-1)} \end{matrix} & \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} \\ \begin{matrix} w_{n1} & \cdots & w_{n(n-1)} \end{matrix} & \begin{matrix} 1 \end{matrix} \end{pmatrix}$$

6.2 Existence of (G^0, G^1, G)

Proposition 6.2 (Existence of G in 3D). $\exists G$ such that the predicate (G^0, G^1, G) is satisfied if and only if

$$\forall (i, j) \in \llbracket 1, 2 \rrbracket^2, \quad \begin{cases} (w_{ij}^1 > 0 \implies w_{ij}^0 > 0) \wedge (w_{ij}^0 \leq 0 \implies w_{ij}^1 \leq 0) \\ (w_{1j}^1 + w_{2j}^1 > 0 \implies w_{1j}^0 + w_{2j}^0 > 0) \wedge (w_{1j}^0 + w_{2j}^0 \leq 0 \implies w_{1j}^1 + w_{2j}^1 \leq 0) \end{cases}$$

Example 6.1 (Counter-example). *Given*

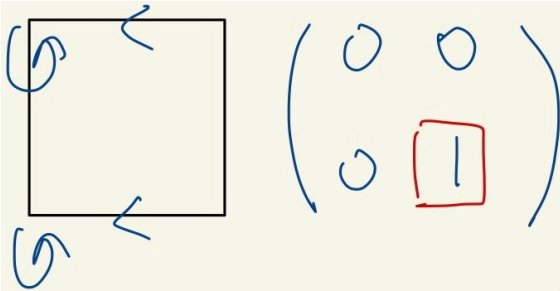
$$W^0 \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W^1 \equiv \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \left(W^1 \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W^0 \equiv \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{Respectively} \right)$$

$\nexists W$ such that their transition graph associated (G^0, G^1, G) verifies the predicate.

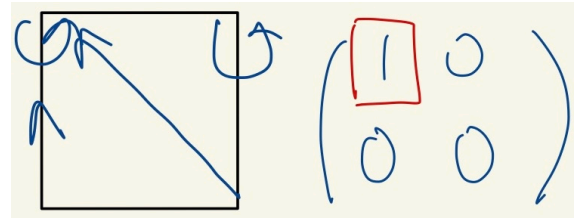
Proof. Taking

$$W^0 \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W^1 \equiv \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad W = \begin{pmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{pmatrix} = \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix}$$

Their transition graph associated is Figure (6.1b) and Figure (6.1a), G should be like Figure



(a) Transition Graph 1



(b) Transition Graph 2

6.2. Observing G we have $E_s(101) = 101$, then applying the technique mentioned in section

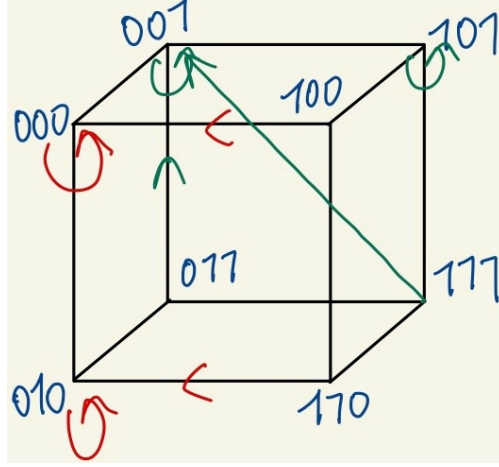


Figure 6.2: Example G

3, we get

$$\begin{aligned}
 \mathbf{x}_s^5 &= \text{sign}((L_1 + L_3)^\top) \quad \text{since } x_1^5 = x_3^5 = 1, x_2^5 = 0 \\
 &= \text{sign} \left(\begin{pmatrix} w_{11} + w_{31} \\ w_{12} + w_{32} \\ w_{13} + w_{33} \end{pmatrix} \right) \\
 &= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad (\text{Closure constraints}) \\
 &\implies \text{sign}(w_{11} + w_{31}) = 1 \\
 &\iff w_{11} + w_{31} > 0
 \end{aligned}$$

The SBN constraints gives $w_{31} \leq 0$, then $w_{11} > 0$. Taking $E = 10\mathbf{0}$, according to Figure 6.2, $E_s(100) = 000$. However,

$$\mathbf{x}_s^1 = \text{sign} \left(\begin{pmatrix} w_{11} \\ w_{12} \\ \mathbf{0} \end{pmatrix} \right) = \begin{pmatrix} \mathbf{1} \\ \text{sign}(w_{12}) \\ \mathbf{0} \end{pmatrix} \neq \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix}$$

Contradiction.

The proof of the case

$$W^1 \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W^0 \equiv \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

is similar. ■

Example 6.2 (Another counter-example). *Given*

$$W_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W_2 = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad \left(W^1 \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W^0 \equiv \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \quad \text{Respectively} \right)$$

$\nexists W$ such that their transition graphs associated (G^0, G^1, G) verify the predicate.

Proof. The proof of $\nexists W$ taking $W^0 \equiv W_1$ and $W^1 \equiv W_2$ is similar as the previous example, we only proof the case $W^0 \equiv W_2$ and $W^1 \equiv W_1$ here.

Taking

$$W^0 \equiv W_2, \quad W^1 \equiv W_1, \quad W = \begin{pmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{pmatrix} = \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix}$$

Their transition graphs associated are Figure (6.3a) and Figure (6.3b), G should be like

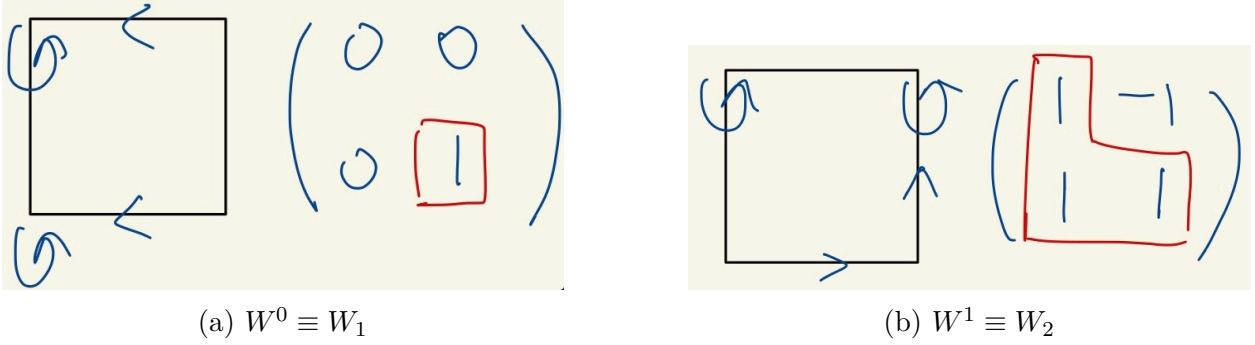


Figure 6.4. Observing G we have $E_s(111) = 011$, then applying the technique mentioned in

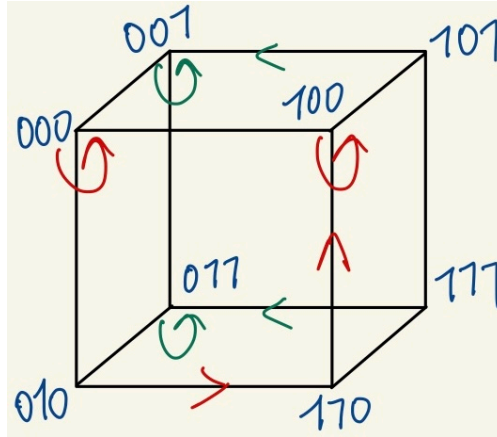


Figure 6.4: Example G3

section 3, we get

$$\begin{aligned}
\mathbf{x}_s^7 &= \text{sign}((L_1 + L_2 + L_3)^\top) \quad \text{since } x_1^7 = x_2^7 = x_3^7 = 1 \\
&= \text{sign} \left(\begin{pmatrix} w_{11} + w_{12} + w_{31} \\ w_{12} + w_{22} + w_{32} \\ w_{13} + w_{23} + w_{33} \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \\
&\implies \text{sign}(w_{12} + w_{22} + w_{32}) = 1 \\
&\iff w_{12} + w_{22} + w_{32} > 0
\end{aligned}$$

The SBN constraints gives $w_{32} \leq 0$, then $w_{12} + w_{22} > 0$. Taking $E = 110$, according to Figure 6.4, $E_s = 100$. However,

$$\mathbf{x}_s^1 = \text{sign} \left(\begin{pmatrix} w_{11} + w_{21} \\ w_{12} + w_{22} \\ 0 \end{pmatrix} \right) = \begin{pmatrix} \text{sign}(w_{11} + w_{12}) \\ 1 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Contradiction. ■

Example 6.3. *Given*

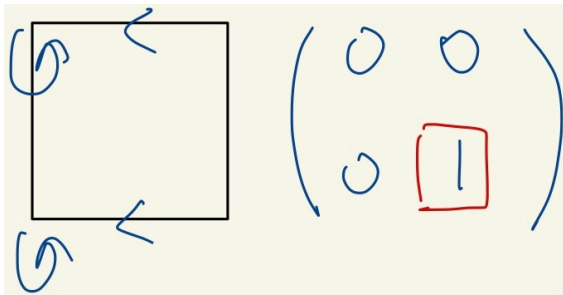
$$W_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

- if we choose $W^0 \equiv W_1$, $W^1 \equiv W_2$, $\exists W$ such that the triplet of their transition graph associated (G^0, G^1, G) verifies the predicate. This case will be proved in the next chapter using composition method.
- if we choose $W^0 \equiv W_2$, $W^1 \equiv W_1$, $\nexists W$ such that the triplet of their transition graph associated (G^0, G^1, G) verifies the predicate.

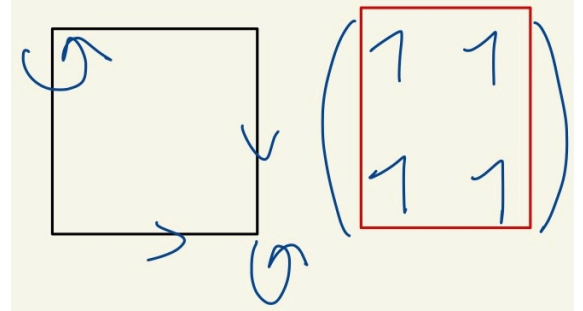
Proof. Here we prove $\nexists W$ such that their transition graphs associated (G^0, G^1, G) verify the predicate. Taking

$$W^0 \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad W^1 \equiv \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad W = \begin{pmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{pmatrix} = \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix}$$

Their transition graphs associated are Figure (6.5a) and Figure (6.5b), G should be like



(a) $W^0 \equiv W_1$



(b) $W^1 \equiv W_2$

Figure 6.6. Observing G we have $E_s(101) = 111$, then applying the technique mentioned in

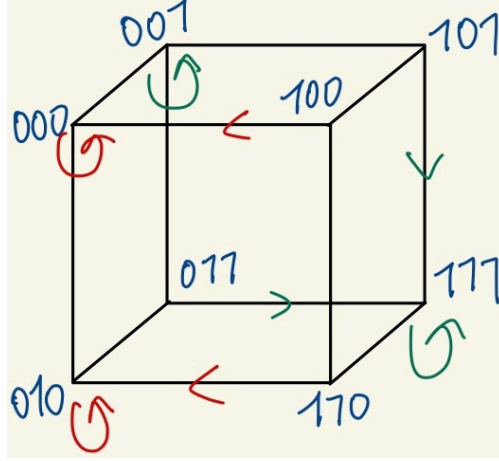


Figure 6.6: Example G3

section 3, we get

$$\begin{aligned}
\mathbf{x}_s^5 &= \text{sign}((L_1 + L_3)^\top) \quad \text{since } x_1^5 = x_3^5 = 1, x_2^5 = 0 \\
&= \text{sign} \left(\begin{pmatrix} w_{11} + w_{31} \\ w_{12} + w_{32} \\ w_{13} + w_{33} \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \\
&\implies \begin{cases} \text{sign}(w_{11} + w_{31}) = 1 \\ \text{sign}(w_{12} + w_{32}) = 1 \end{cases} \\
&\iff \begin{cases} w_{11} + w_{31} > 0 \\ w_{12} + w_{32} > 0 \end{cases}
\end{aligned}$$

The SBN constraints gives $w_{31} \leq 0$, $w_{32} \leq 0$, then $w_{11} > 0$, $w_{12} > 0$. Taking $E = 100$, according to Figure 6.4, $E_s = 000$. However,

$$\mathbf{x}_s^1 = \text{sign} \left(\begin{pmatrix} w_{11} \\ w_{12} \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Contradiction. ■

Proposition 6.3 (Existence of G in nD). $\exists G$ such that the predicate (G^0, G^1, G) is satisfied if and only if $\forall (i, j) \in \llbracket 1, n-1 \rrbracket^2, \mathbf{x}^E \in \mathcal{B}^n$,

$$\left(\sum_{i=1}^{n-1} w_{ij}^1 x_i^E > 0 \implies \sum_{i=1}^{n-1} w_{ij}^0 x_i^E > 0 \right) \wedge \left(\sum_{i=1}^{n-1} w_{ij}^0 x_i^E \leq 0 \implies \sum_{i=1}^{n-1} w_{ij}^1 x_i^E \leq 0 \right) \quad (6.5)$$

Proof. The proof will proceed by contradiction. Assume that $\exists G$ such that the predicate

(G^0, G^1, G) hold true and also satisfy

$$\neg \left(\sum_{i=1}^{n-1} w_{ij}^1 x_i^E > 0 \implies \sum_{i=1}^{n-1} w_{ij}^0 x_i^E > 0 \right) \vee \neg \left(\sum_{i=1}^{n-1} w_{ij}^0 x_i^E \leq 0 \implies \sum_{i=1}^{n-1} w_{ij}^1 x_i^E \leq 0 \right)$$

which means $\exists \mathbf{y}^E \in \mathcal{B}^n$ such that

$$\left(\sum_{i=1}^{n-1} w_{ij}^1 y_i^E > 0 \implies \sum_{i=1}^{n-1} w_{ij}^0 y_i^E \leq 0 \right) \vee \left(\sum_{i=1}^{n-1} w_{ij}^0 y_i^E \leq 0 \implies \sum_{i=1}^{n-1} w_{ij}^1 y_i^E > 0 \right) \quad (6.6)$$

By Properties 6.1 and 6.2, $\forall (i, j) \in \llbracket 1, n-1 \rrbracket, \mathbf{x}^E \in \mathcal{B}^n$, we have

$$\begin{aligned} \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i^0 x_i^E \right)^\top \right) &= \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i x_i^E \right)^\top \right) \\ \iff \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^0 x_i^E \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij} x_i^E \right) \\ \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i^1 x_i^E \right)^\top \right) &= \text{sign} \left(\left(\sum_{i=1}^{n-1} L_i x_i^E + L_n \right)^\top \right) \\ \iff \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^1 x_i^E \right) &= \text{sign} \left(\sum_{i=1}^{n-1} w_{ij} x_i^E + w_{nj} \right) \end{aligned}$$

Then

$$\begin{aligned} \sum_{i=1}^{n-1} w_{ij}^1 x_i^E > 0 &\iff \sum_{i=1}^{n-1} w_{ij} x_i^E + w_{nj} > 0 \\ &\implies \sum_{i=1}^{n-1} w_{ij} x_i^E > 0 \quad \text{since } w_{nj} \leq 0 \text{ given by the SBN constraint 5.7} \\ &\implies \sum_{i=1}^{n-1} w_{ij}^0 x_i^E > 0 \text{ according to the Property 6.1} \end{aligned}$$

Consequently,

$$\forall (i, j) \in \llbracket 1, n-1 \rrbracket, \forall \mathbf{x}^E \in \mathcal{B}^n \quad \sum_{i=1}^{n-1} w_{ij}^1 x_i^E > 0 \implies \sum_{i=1}^{n-1} w_{ij}^0 x_i^E > 0$$

Contradiction. Similarly,

$$\begin{aligned}
\sum_{i=1}^{n-1} w_{ij}^0 x_i^E \leq 0 &\iff \sum_{i=1}^{n-1} w_{ij} x_i^E \leq 0 \\
&\implies \sum_{i=1}^{n-1} w_{ij} x_i^E + w_{nj} \leq 0 \\
&\implies \sum_{i=1}^{n-1} w_{ij}^1 x_i^E \leq 0
\end{aligned}$$

$$\forall (i, j) \in \llbracket 1, n-1 \rrbracket, \forall \mathbf{x}^E \in \mathcal{B}^n \quad \sum_{i=1}^{n-1} w_{ij}^0 x_i^E \leq 0 \implies \sum_{i=1}^{n-1} w_{ij}^1 x_i^E \leq 0$$

Contradiction. Thus, $\nexists \mathbf{y}^E \in \mathcal{B}^n$ that satisfies the specified conditions 6.6, meaning $\exists G$ such that the predicate (G^0, G^1, G) is satisfied if and only if $\forall (i, j) \in \llbracket 1, n-1 \rrbracket^2, \mathbf{x}^E \in \mathcal{B}^n$,

$$\left(\sum_{i=1}^{n-1} w_{ij}^1 x_i^E > 0 \implies \sum_{i=1}^{n-1} w_{ij}^0 x_i^E > 0 \right) \wedge \left(\sum_{i=1}^{n-1} w_{ij}^0 x_i^E \leq 0 \implies \sum_{i=1}^{n-1} w_{ij}^1 x_i^E \leq 0 \right)$$

■

Chapter 7

Composition method

According to the previous section, we propose the following method to construct the weight matrix W using two sub weight matrices W^0 and W^1 , whose existence of W is verified.

Algorithm 1: Composition Method

Data: W_{init}^0 and W_{init}^1 2 weight matrix associate with G^0 and G^1 2 TGs

Result: W the weight matrix of the composition TG G

```

1  $W \leftarrow \begin{pmatrix} W^0 & \mathbf{0} \in \mathbb{O}^{n-1} \\ \mathbf{0} \in \mathbb{O}^{n-1} & 1 \end{pmatrix}$ 
2 for  $E$  in  $G^1 \setminus \{00 \dots 01\}$  do
3   if  $sign \circ f(\mathbf{x}^E) \neq \mathbf{x}_s^E$  then
4     modify  $W$  such that  $sign \circ f(\mathbf{x}^E) = \mathbf{x}_s^E$ 
5     if  $W^0 \not\equiv W|_{dim=n-1}$  then
6       modify  $W|_{dim=n-1}$  such that  $W^0 \equiv W|_{dim=n-1}$ 
7 end
```

Example 7.1. We will illustrate the process of Algorithm 1 using example 6.3. Taking

$$W^0 \equiv \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad W^1 \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

G should resemble the structure depicted in Figure 7.1.

1. Construct matrix W based on W^0 by incorporating the basic SBN constraints and closure constraints as defined in Property 6.1, then

$$W^{Step1} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

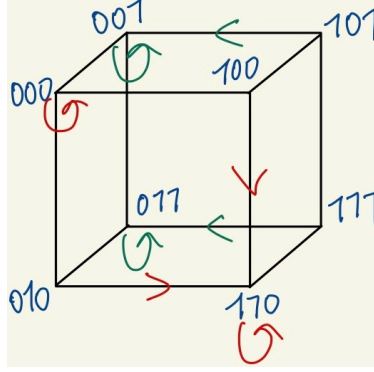


Figure 7.1: Objective G

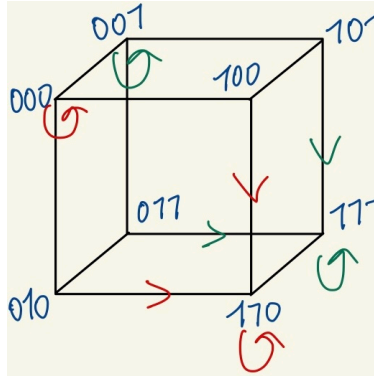


Figure 7.2: Step1

2. $E_s(101) = 001$,

$$\mathbf{x}_s^5 = \text{sign}((L_1^{\text{Step1}} + L_3^{\text{Step1}})^\top) = \text{sign} \left(\begin{pmatrix} 1+0 \\ 1+0 \\ 0+1 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

then modify L_3 from $(0 \ 0 \ 1)$ to $(-1 \ -1 \ 1)$ such that

$$\mathbf{x}_s^5 = \text{sign}((L_1^{\text{Step2}} + L_3^{\text{Step2}})^\top) = \text{sign} \left(\begin{pmatrix} 1-1 \\ 1-1 \\ 0+1 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Now

$$W^{\text{Step2}} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$$

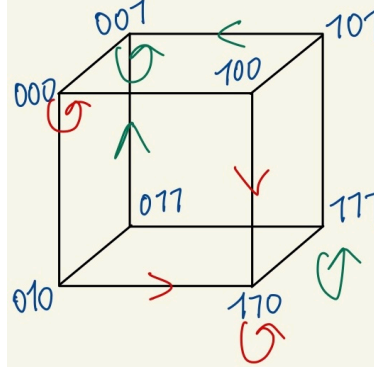


Figure 7.3: Step2

3. $E_s(011) = 011$,

$$\mathbf{x}_s^6 = \text{sign}((L_2^{\text{Step2}} + L_3^{\text{Step2}})^\top) = \text{sign} \left(\begin{pmatrix} 1-1 \\ 1-1 \\ 0+1 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

then modify L_2^{Step2} from $(1 \ 1 \ 0)$ to $(1 \ 2 \ 0)$ such that

$$\mathbf{x}_s^6 = \text{sign}((L_2^{\text{Step3}} + L_3^{\text{Step3}})^\top) = \text{sign} \left(\begin{pmatrix} 1-1 \\ 2-1 \\ 0+1 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Now

$$W^{\text{Step3}} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ -1 & -1 & 1 \end{pmatrix}$$

Since L_2^{Step3} are modified, we must verify $W^{\text{Step3}}|_{\dim=2} \equiv W^0$:

$$\text{sign}(L_1^{\text{Step3}\top}) = \text{sign}(L_2^{\text{Step3}\top}) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \text{sign}(L_1^0) = \text{sign}(L_2^0)$$

$$\text{sign}((L_1^{\text{Step3}} + L_2^{\text{Step3}})^\top) = \text{sign} \left(\begin{pmatrix} 1+1 \\ 1+2 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \text{sign}(L_1^0 + L_2^0)$$

4. $E_s(111) = 011$,

$$\mathbf{x}_s^7 = \text{sign}((L_1^{\text{Step3}} + L_2^{\text{Step3}} + L_3^{\text{Step3}})^\top) = \text{sign} \left(\begin{pmatrix} 1+1-1 \\ 1+2-1 \\ 0+0+1 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

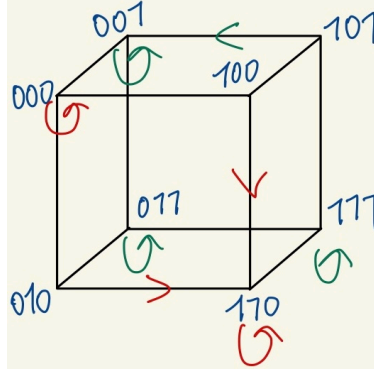


Figure 7.4: Step3

then modify L_3^{Step3} from $(-1 \ -1 \ 1)$ to $(-2 \ -1 \ 1)$ such that

$$\mathbf{x}_s^7 = \text{sign}((L_1^{Step4} + L_2^{Step4} + L_3^{Step4})^\top) = \text{sign}\left(\begin{pmatrix} 1 + 1 - 2 \\ 1 + 2 - 1 \\ 0 + 0 + 1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Now

$$W^{Step4} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ -2 & -1 & 1 \end{pmatrix}$$

We thereby obtain the objective G .

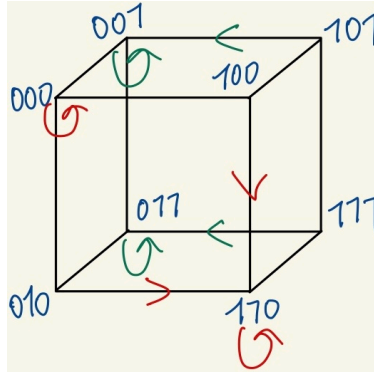


Figure 7.5: Step4

Remark 7.0.1. The non-existence of W can be proven if the algorithm returns a matrix equivalent to W^{Step1} or falls into an infinite loop.

Part III

Constraints, Properties of weight matrices (2 control bits)

Chapter 8

SBN Constraints

Applying the same approach as in the case with one control bit, this chapter will explore the SBN Constraints for two control bits.

In the subsequent content, we will use x_{n-1} and x_n as the control bits to simplify representations. However, the method is applicable to any control bits tuple (x_i, x_j) , where $(i, j) \in \llbracket 1, n \rrbracket^2$, $i < j$ with minor modifications.

8.1 SBN Constraints 3D

In 3D, $\forall k \in \{0, 1\}$, G^k associates with R^k is SBN, where $k = k_1 \oplus k_2$ and G^k has the same behaviors as G gives

$$\begin{aligned} \forall x_1 \in \mathcal{B}, \mathbf{x}_s^E = \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ k_1 \\ k_2 \end{pmatrix} \right) &= \text{sign} \circ f_1^k(x_1, \text{sign}(k_1 + (1 - 2k)k_2)) \\ \Leftrightarrow \left\{ \begin{array}{l} \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ 0 \\ 0 \end{pmatrix} \right) &= \text{sign} \circ f_1^0(x_1, 0) \\ \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ 1 \\ 1 \end{pmatrix} \right) &= \text{sign} \circ f_1^0(x_1, 1) \\ \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ 0 \\ 1 \end{pmatrix} \right) &= \text{sign} \circ f_1^1(x_1, 0) \\ \text{sign} \circ f_1 \left(\begin{pmatrix} x_1 \\ 1 \\ 0 \end{pmatrix} \right) &= \text{sign} \circ f_1^1(x_1, 1) \end{array} \right. \end{aligned}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for both $k = 0$ and $k = 1$. Consequently, we examine their intersection:

$$\begin{aligned}
& (sign(w_{11}x_1) = sign(w_{11}^0x_1)) \\
& \wedge (sign(w_{11}x_1 + w_{21} + w_{31}) = sign(w_{11}^0x_1 + w_{21}^0)) \\
& \wedge (sign(w_{11}x_1 + w_{31}) = sign(w_{11}^1x_1)) \\
& \wedge (sign(w_{11}x_1 + w_{21}) = sign(w_{11}^1x_1 + w_{21}^1))
\end{aligned}$$

Enumerating all possible values of tuple x_1 to more precisely define the constraints:

- $x_1 = 0$:

$$(sign(0) = sign(0)) \wedge (sign(w_{21} + w_{31}) = sign(w_{21}^0)) \quad (8.1)$$

$$\begin{aligned}
& (sign(w_{31}) = sign(0)) \wedge (sign(w_{21}) = sign(w_{21}^1)) \\
& \implies (w_{31} \leq 0) \wedge (sign(w_{21}) = sign(w_{21}^1))
\end{aligned} \quad (8.2)$$

- $x_1 = 1$:

$$(sign(w_{11}) = sign(w_{11}^0)) \wedge (sign(w_{11} + w_{21} + w_{31}) = sign(w_{11}^0 + w_{21}^0)) \quad (8.3)$$

$$(sign(w_{11} + w_{31}) = sign(w_{11}^1)) \wedge (sign(w_{21}) = sign(w_{11}^1 + w_{21}^1)) \quad (8.4)$$

By combining Constraint (8.1), (8.2), (8.3) and Constraint (8.4), we can derive the SBN constraints 2 control bits for 3D:

$$\begin{cases}
w_{31} \leq 0 \\
sign(w_{11}) = sign(w_{11}^0) \\
sing(w_{21}) = sign(w_{21}^1) \\
sign(w_{11} + w_{21}) = sign(w_{11}^1 + w_{21}^1) \\
sign(w_{11} + w_{31}) = sign(w_{11}^1) \\
sign(w_{21} + w_{31}) = sign(w_{21}^0) \\
sign(w_{11} + w_{21} + w_{31}) = sign(w_{11}^0 + w_{21}^0)
\end{cases}$$

8.2 SBN Constraints 4D

In 4D, $\forall k \in \{0, 1\}$, G^k associates with R^k is SBN, where $k = k_1 \oplus k_2$ and G^k has the same behaviors as G gives

$$\begin{aligned} \forall (x_1, x_2) \in \mathcal{B}^2, \mathbf{x}_s^E = \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ k_1 \\ k_2 \end{pmatrix} \right) &= \text{sign} \circ f_i^k(x_1, x_2, \text{sign}(k_1 + (1 - 2k)k_2)) \\ \Leftrightarrow \left\{ \begin{aligned} \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{pmatrix} \right) &= \text{sign} \circ f_i^0(x_1, x_2, 0) \\ \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \\ 1 \end{pmatrix} \right) &= \text{sign} \circ f_i^0(x_1, x_2, 1) \\ \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \\ 1 \end{pmatrix} \right) &= \text{sign} \circ f_i^1(x_1, x_2, 0) \\ \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \\ 0 \end{pmatrix} \right) &= \text{sign} \circ f_i^1(x_1, x_2, 1) \end{aligned} \right. \end{aligned}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for both $k = 0$ and $k = 1$. Consequently, we examine their intersection:

$$\begin{aligned} &(\text{sign}(w_{1i}x_1 + w_{2i}x_2) = \text{sign}(w_{1i}^0x_1 + w_{2i}^0x_2)) \\ &\wedge (\text{sign}(w_{1i}x_1 + w_{2i}x_2 + w_{3i} + w_{4i}) = \text{sign}(w_{1i}^0x_1 + w_{2i}^0x_2 + w_{3i}^0)) \\ &\wedge (\text{sign}(w_{1i}x_1 + w_{2i}x_2 + w_{3i}) = \text{sign}(w_{1i}^1x_1 + w_{2i}^1x_2 + w_{3i}^1)) \\ &\wedge (\text{sign}(w_{1i}x_1 + w_{2i}x_2 + w_{4i}) = \text{sign}(w_{1i}^1x_1 + w_{2i}^1x_2)) \end{aligned}$$

Enumerating all possible values of tuple (x_1, x_2) to more precisely define the constraints:

- $x_1 = x_2 = 0$:

$$(\text{sign}(0) = \text{sign}(0)) \wedge (\text{sign}(w_{3i} + w_{4i}) = \text{sign}(w_{3i}^0)) \quad (8.5)$$

$$\begin{aligned} &(\text{sign}(w_{3i}) = \text{sign}(w_{3i}^1)) \wedge (\text{sign}(w_{4i}) = \text{sign}(0)) \\ &\implies (w_{4i} \leq 0) \wedge (\text{sign}(w_{3i}) = \text{sign}(w_{3i}^1)) \end{aligned} \quad (8.6)$$

- $x_1 = 1, x_2 = 0$:

$$\begin{aligned} & (\text{sign}(w_{1i}) = \text{sign}(w_{1i}^0)) \\ & \wedge (\text{sign}(w_{1i} + w_{3i} + w_{4i}) = \text{sign}(w_{1i}^0 + w_{3i}^0)) \end{aligned} \quad (8.7)$$

$$\begin{aligned} & (\text{sign}(w_{1i} + w_{3i}) = \text{sign}(w_{1i}^1 + w_{3i}^1)) \\ & \wedge (\text{sign}(w_{1i} + w_{4i}) = \text{sign}(w_{1i}^1)) \end{aligned} \quad (8.8)$$

- $x_1 = 0, x_2 = 1$:

$$\begin{aligned} & (\text{sign}(w_{2i}) = \text{sign}(w_{2i}^0)) \\ & \wedge (\text{sign}(w_{2i} + w_{3i} + w_{4i}) = \text{sign}(w_{2i}^0 + w_{3i}^0)) \end{aligned} \quad (8.9)$$

$$\begin{aligned} & (\text{sign}(w_{2i} + w_{3i}) = \text{sign}(w_{2i}^1 + w_{3i}^1)) \\ & \wedge (\text{sign}(w_{2i} + w_{4i}) = \text{sign}(w_{2i}^1)) \end{aligned} \quad (8.10)$$

- $x_1 = 1, x_2 = 1$:

$$\begin{aligned} & (\text{sign}(w_{1i} + w_{2i}) = \text{sign}(w_{1i}^0 + w_{2i}^0)) \\ & \wedge (\text{sign}(w_{1i} + w_{2i} + w_{3i} + w_{4i}) = \text{sign}(w_{1i}^0 + w_{2i}^0 + w_{3i}^0)) \end{aligned} \quad (8.11)$$

$$\begin{aligned} & (\text{sign}(w_{1i} + w_{2i} + w_{3i}) = \text{sign}(w_{1i}^1 + w_{2i}^1 + w_{3i}^1)) \\ & \wedge (\text{sign}(w_{1i} + w_{2i} + w_{4i}) = \text{sign}(w_{1i}^1 + w_{2i}^1)) \end{aligned} \quad (8.12)$$

By synthesizing the 8 listed Constraints (8.5, 8.6, 8.7, 8.8, 8.9, 8.10, 8.11, 8.12), we can derive the SBN constraints for two control bits in a 4D setting.

$$\left\{ \begin{array}{l} w_{4i} \leq 0 \\ \text{sign}(w_{1i}) = \text{sign}(w_{1i}^0) \\ \text{sign}(w_{2i}) = \text{sign}(w_{2i}^0) \\ \text{sign}(w_{3i}) = \text{sign}(w_{3i}^1) \\ \text{sign}(w_{1i} + w_{2i}) = \text{sign}(w_{1i}^0 + w_{2i}^0) \\ \text{sign}(w_{1i} + w_{3i}) = \text{sign}(w_{1i}^1 + w_{3i}^1) \\ \text{sign}(w_{1i} + w_{4i}) = \text{sign}(w_{1i}^1) \\ \text{sign}(w_{2i} + w_{4i}) = \text{sign}(w_{2i}^1) \\ \text{sign}(w_{3i} + w_{4i}) = \text{sign}(w_{3i}^0) \\ \text{sign}(w_{1i} + w_{2i} + w_{3i}) = \text{sign}(w_{1i}^1 + w_{2i}^1 + w_{3i}^1) \\ \text{sign}(w_{1i} + w_{2i} + w_{4i}) = \text{sign}(w_{1i}^1 + w_{2i}^1) \\ \text{sign}(w_{1i} + w_{3i} + w_{4i}) = \text{sign}(w_{1i}^0 + w_{3i}^0) \\ \text{sign}(w_{2i} + w_{3i} + w_{4i}) = \text{sign}(w_{2i}^0 + w_{3i}^0) \\ \text{sign}(w_{1i} + w_{2i} + w_{3i} + w_{4i}) = \text{sign}(w_{1i}^0 + w_{2i}^0 + w_{3i}^0) \end{array} \right.$$

8.3 Generalized SBN Constraints

Proposition 8.1 (Generalized SBN Constraints for 2 control bits). $\forall j \in \llbracket 1, n-1 \rrbracket, \forall \mathbf{x}^E \in \mathcal{B}^n$,

$$\text{sign} \left(\sum_{i=1}^n w_{ij} x_i \right) = \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^k x_i + w_{(n-1)j}^k \cdot \text{sign}(x_{n-1} + (1-2k)x_n) \right) \quad (8.13)$$

where $k = x_{n-1} \oplus x_n$. In particular, we have

$$w_{nj} \leq 0 \quad (8.14)$$

$$\text{sign}(w_{(n-1)j}) = \text{sign}(w_{(n-1)j}^1) \quad (8.15)$$

$$\text{sign}(w_{(n-1)j} + w_{nj}) = \text{sign}(w_{(n-1)j}^0) \quad (8.16)$$

Proof. Similarly as in 3D and 4D, G^k have the same behaviors as G gives

$$\forall (x_1, \dots, x_{n-2}) \in \mathcal{B}^{n-2}, \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ \vdots \\ x_{n-2} \\ k_1 \\ k_2 \end{pmatrix} \right) = \text{sign} \circ f_i^k \left(\begin{pmatrix} x_1 \\ \vdots \\ x_{n-2} \\ \text{sign}(k_1 + (1-2k)k_2) \end{pmatrix} \right) \quad (8.17)$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above $\forall k = k_1 \oplus k_2 \in \{0, 1\}$. Consequently, we examine their intersection:

$$\text{sign} \left(\sum_{i=1}^{n-2} w_{ij} x_i + w_{(n-1)j} k_1 + w_{nj} k_2 \right) = \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^k x_i + w_{(n-1)j}^k \text{sign}(k_1 + (1-2k)k_2) \right)$$

Note that $l = \text{sign}(k_1 + (1-2k)k_2)$, S the set of $(x_1, \dots, x_{n-2})$ whose value is 1 and $\#S$ the cardinal of the set S , then:

- $\#S = 0 \iff x_1 = \dots = x_{n-2} = 0$:

$$\bigwedge_{j=1}^{n-1} \text{sign} \left(\sum_{i=1}^{n-1} w_{ij} \cdot 0 + w_{(n-1)j} k_1 + w_{nj} k_2 \right) = \text{sign} \left(\sum_{i=1}^{n-1} w_{ij}^k \cdot 0 + w_{(n-1)j}^k l \right)$$

$$\bigwedge_{j=1}^{n-1} \text{sign}(w_{(n-1)j} k_1 + w_{nj} k_2) = \text{sign}(w_{(n-1)j}^k l) \quad \forall (k_1, k_2) \in \mathcal{B}^2$$

$$\iff \begin{cases} \text{sign}(w_{nj}) = 0 \implies w_{nj} \leq 0 \\ \text{sign}(w_{(n-1)j} + w_{nj}) = \text{sign}(w_{(n-1)j}^0) \\ \text{sign}(w_{(n-1)j}) = \text{sign}(w_{(n-1)j}^1) \end{cases} \quad (8.18)$$

- $\#S = 1 \iff x_a = 1, \forall b \neq a, x_b = 0 \iff \forall (a, j) \in \llbracket 1, n-2 \rrbracket^2, (k_1, k_2) \in \mathcal{B}^2$:

$$\begin{aligned} \text{sign} \left(\sum_{i=1, i \neq a}^{n-1} w_{ij} \cdot 0 + w_{aj} + w_{(n-1)j}k_1 + w_{nj}k_2 \right) &= \text{sign} \left(\sum_{i=1, i \neq a}^{n-2} w_{ij}^k \cdot 0 + w_{aj}^k + w_{(n-1)j}^k l \right) \\ \text{sign} (w_{aj} + w_{(n-1)j}k_1 + w_{nj}k_2) &= \text{sign}(w_{aj}^k + w_{(n-1)j}^k l) \end{aligned} \quad (8.19)$$

- $\#S = 2, \dots, \#S = n-2$:

$$\vdots \quad (8.20)$$

- $\#S = n-1 \iff x_1 = \dots = x_{n-1} = 1$:

$$\begin{aligned} \bigwedge_{j=1}^{n-2} \text{sign} \left(\sum_{i=1}^{n-2} w_{ij} \cdot 1 + w_{(n-1)j}k_1 + w_{nj}k_2 \right) &= \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^k \cdot 1 + w_{(n-1)j}^k l \right) \\ \bigwedge_{j=1}^{n-1} \text{sign} \left(\sum_{i=1}^{n-2} w_{ij} + w_{(n-1)j}k_1 + w_{nj}k_2 \right) &= \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^k + w_{(n-1)j}^k l \right) \end{aligned} \quad (8.21)$$

By combining Constraint (8.18), (8.19), (8.20) and (8.21), we can derive the SBN constraints for n dimensions: $\forall (x_1, \dots, x_{n-2}) \in \mathcal{B}^{n-2}$,

$$\begin{aligned} &\bigwedge_{(k_1, k_2) \in \mathcal{B}^2} \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}x_i + w_{(n-1)j}k_1 + w_{nj}k_2 \right) = \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^k x_i + w_{(n-1)j}^k l \right) \\ \iff &\forall \mathbf{x}^E \in \mathcal{B}^n, \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}x_i + w_{(n-1)j}x_{n-1} + w_{nj}x_n \right) \\ &= \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^k x_i + w_{(n-1)j}^k \text{sign}(x_{n-1} + (1-2k)x_n) \right) \\ \iff &\begin{cases} \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}x_i \right) = \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^0 x_i \right) \\ \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}x_i + w_{nj} \right) = \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^1 x_i \right) \\ \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}x_i + w_{(n-1)j} \right) = \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^1 x_i + w_{(n-1)j}^1 \right) \\ \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}x_i + w_{(n-1)j} + w_{nj} \right) = \text{sign} \left(\sum_{i=1}^{n-2} w_{ij}^0 x_i + w_{(n-1)j}^0 \right) \end{cases} \end{aligned} \quad (8.22)$$

■

Hypothesis 8.1. *In the general case, where the two control bits are not necessarily (x_{n-1}, x_n) but any pair (x_i, x_j) with $(i, j) \in \llbracket 1, n \rrbracket$, $i < j$, the formula 8.17 presented in the proof needs to be adjusted as follows:*

$$\forall \mathbf{x}^{\mathbf{E}} \in \mathcal{B}^n, \text{sign} \circ f_i \left(\begin{pmatrix} x_1 \\ \vdots \\ x_i = k_1 \\ \vdots \\ x_j = k_2 \\ \vdots \\ x_n \end{pmatrix} \right) = \text{sign} \circ f_i^k \left(\begin{pmatrix} x_1 \notin \{x_i, x_j\} \\ \vdots \\ x_n \notin \{x_i, x_j\} \\ \text{sign}(k_1 + (1 - 2k)k_2) \end{pmatrix} \right)$$

Chapter 9

Closure Constraints

As in the previous chapter, we will focus on discussing the Closure Constraints for two control bits, simplifying our representations by fixing (x_{n-1}, x_n) as the control bit tuple and following the same approach used for a single control bit.

9.1 Closure constraints 3D

In 3D, the definition of closure implies that the second bit and the third bit of each state E satisfy the following conditions:

$$\begin{aligned} \forall \mathbf{x}^E \in \mathcal{B}^3, \quad G^0 &= \{sign \circ f_2(\mathbf{x}^E) = sign \circ f_3(\mathbf{x}^E)\} \\ &= \{sign \circ f_2(\mathbf{x}^E) - sign \circ f_3(\mathbf{x}^E) = 0\} \\ G^1 &= \{sign \circ f_2(\mathbf{x}^E) = \neg sign \circ f_3(\mathbf{x}^E)\} \\ &= \{sign \circ f_2(\mathbf{x}^E) + sign \circ f_3(\mathbf{x}^E) = 1\} \end{aligned}$$

which is equivalent to $\forall x_1 \in \mathcal{B}, k = k_1 \oplus k_2$,

$$sign \circ f_2(\mathbf{x}^E) + (2k - 1)sign \circ f_3(\mathbf{x}^E) = k$$

$$\iff \begin{cases} f_2 \left(\begin{pmatrix} x_1 \\ 0 \\ 0 \end{pmatrix} \right) - sign \circ f_3 \left(\begin{pmatrix} x_1 \\ 0 \\ 0 \end{pmatrix} \right) = f_2 \left(\begin{pmatrix} x_1 \\ 1 \\ 1 \end{pmatrix} \right) - sign \circ f_3 \left(\begin{pmatrix} x_1 \\ 1 \\ 1 \end{pmatrix} \right) = 0 \\ f_2 \left(\begin{pmatrix} x_1 \\ 0 \\ 1 \end{pmatrix} \right) + sign \circ f_3 \left(\begin{pmatrix} x_1 \\ 0 \\ 1 \end{pmatrix} \right) = f_2 \left(\begin{pmatrix} x_1 \\ 1 \\ 0 \end{pmatrix} \right) + sign \circ f_3 \left(\begin{pmatrix} x_1 \\ 1 \\ 0 \end{pmatrix} \right) = 1 \end{cases}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for all $k \in \mathcal{B}$. Consequently, we examine their intersection:

- $k = k_1 \oplus k_2 = 0$:

$$(sign(w_{12}x_1) = sign(w_{13}x_1)) \wedge (sign(w_{12}x_1 + w_{22} + w_{32}) = sign(w_{13}x_1 + w_{23} + w_{33}))$$

- $k = k_1 \oplus k_2 = 1$:

$$(sign(w_{12}x_1 + w_{22}) + sign(w_{13}x_1 + w_{23}) = 1) \wedge (sign(w_{12}x_1 + w_{32}) + sign(w_{13}x_1 + w_{33}) = 1)$$

Enumerating all possible values of x_1 to more precisely define the constraints:

- Let $x_1 = 0$:

$$(0 = 0) \wedge (sign(w_{22} + w_{32}) = sign(w_{23} + w_{33})) \\ \wedge (sign(w_{22}) + sign(w_{23}) = 1) \wedge (sign(w_{32}) + sign(w_{33}) = 1)$$

which is equivalent to

$$\left\{ \begin{array}{l} ((w_{22} + w_{32} > 0) \wedge (w_{23} + w_{33} > 0)) \vee ((w_{22} + w_{32} \leq 0) \wedge (w_{23} + w_{33} \leq 0)) \\ ((w_{22} > 0) \wedge (w_{23} \leq 0)) \vee ((w_{22} \leq 0) \wedge (w_{23} > 0)) \\ ((w_{32} > 0) \wedge (w_{33} \leq 0)) \vee ((w_{32} \leq 0) \wedge (w_{33} > 0)) \end{array} \right. \quad (9.1)$$

We can deduce the following relationships by contradiction:

$$sign(w_{22}) = sign(w_{33}) = 1 - sign(w_{23}) = 1 - sign(w_{32}) \quad (9.2)$$

Assume

$$\left\{ \begin{array}{l} (w_{22} > 0) \wedge (w_{23} \leq 0) \\ (w_{32} > 0) \wedge (w_{33} \leq 0) \end{array} \right. \implies w_{22} + w_{32} > 0$$

However, according to Constraint 9.1, we have $w_{23} + w_{33} \leq 0$. This leads to a contradiction. Similarly, assuming that

$$\left\{ \begin{array}{l} (w_{22} \leq 0) \wedge (w_{23} > 0) \\ (w_{32} \leq 0) \wedge (w_{33} > 0) \end{array} \right. \implies w_{23} + w_{33} > 0$$

This assumption contradicts the statement given by Constraint 9.1 that $w_{23} + w_{33} \leq 0$. Therefore, we can deduce that Constraint 9.1 is equivalent to Constraint 9.2. The matrix representation $sign(W)$ is illustrated in Figure 9.1.

- Let $x_1 = 1$:

$$(sign(w_{12}) = sign(w_{13})) \\ \wedge (sign(w_{12} + w_{22} + w_{32}) = sign(w_{13} + w_{23} + w_{33})) \\ \wedge (sign(w_{12} + w_{22}) + sign(w_{13} + w_{23}) = 1) \\ \wedge (sign(w_{12} + w_{32}) + sign(w_{13} + w_{33}) = 1)$$

$\text{sign}(w_{12})$	$\text{sign}(w_{13})$	$\frac{1}{0}$	$\frac{1}{0}$	$\frac{1}{0}$	$\frac{1}{0}$
$\text{sign}(w_{22})$	$\text{sign}(w_{23})$	1	0	0	1
$\text{sign}(w_{32})$	$\text{sign}(w_{33})$	0	1	1	0

Figure 9.1: Representation of matrix $\text{sign}(W)$

which is equivalent to

$$\left\{ \begin{array}{l} ((w_{12} > 0) \wedge (w_{13} > 0)) \vee ((w_{12} \leq 0) \wedge (w_{13} \leq 0)) \\ ((w_{12} + w_{22} + w_{32} > 0) \wedge (w_{13} + w_{23} + w_{33} > 0)) \\ \vee ((w_{12} + w_{22} + w_{32} \leq 0) \wedge (w_{12} + w_{23} + w_{33} \leq 0)) \\ ((w_{12} + w_{22} > 0) \wedge (w_{13} + w_{23} \leq 0)) \vee ((w_{12} + w_{22} \leq 0) \wedge (w_{13} + w_{23} > 0)) \\ ((w_{12} + w_{32} > 0) \wedge (w_{13} + w_{33} \leq 0)) \vee ((w_{12} + w_{32} \leq 0) \wedge (w_{13} + w_{33} > 0)) \end{array} \right. \quad (9.3)$$

Similar to the previous case, the proof is constructed based on contradiction. Assume

$$(w_{12} + w_{22} > 0) \wedge (w_{12} + w_{32} > 0)$$

According to Constraint 9.3, we have

$$(w_{13} + w_{23} \leq 0) \wedge (w_{13} + w_{33} \leq 0)$$

Given Constraint 9.2, which states $\text{sign}(w_{22}) = \text{sign}(w_{33}) = 1 - \text{sign}(w_{23}) = 1 - \text{sign}(w_{32})$, we can consider the following two cases:

- 1st case: $(w_{22} > 0) \wedge (w_{33} > 0) \wedge (w_{23} \leq 0) \wedge (w_{32} \leq 0)$

$$(w_{12} > 0) \wedge (w_{13} > 0) \wedge (w_{22} > 0) \wedge (w_{33} > 0) \implies w_{13} + w_{33} > 0 \quad \text{Contradiction}$$

$$(w_{12} \leq 0) \wedge (w_{13} \leq 0) \wedge (w_{23} \leq 0) \wedge (w_{32} \leq 0) \implies w_{12} + w_{32} \leq 0 \quad \text{Contradiction}$$

- 2nd case: $(w_{22} \leq 0) \wedge (w_{33} \leq 0) \wedge (w_{23} > 0) \wedge (w_{32} > 0)$

$$(w_{12} > 0) \wedge (w_{13} > 0) \wedge (w_{23} > 0) \wedge (w_{32} > 0) \implies w_{13} + w_{23} > 0 \quad \text{Contradiction}$$

$$(w_{12} \leq 0) \wedge (w_{13} \leq 0) \wedge (w_{22} \leq 0) \wedge (w_{33} \leq 0) \implies w_{12} + w_{22} \leq 0 \quad \text{Contradiction}$$

Just like the previous scenario, assuming

$$(w_{12} + w_{22} \leq 0) \wedge (w_{12} + w_{32} \leq 0)$$

From Constraint 9.3, we derive:

$$(w_{13} + w_{23} > 0) \wedge (w_{13} + w_{33} > 0)$$

- 1st case: $(w_{22} > 0) \wedge (w_{33} > 0) \wedge (w_{23} \leq 0) \wedge (w_{32} \leq 0)$

$$(w_{12} > 0) \wedge (w_{13} > 0) \wedge (w_{22} > 0) \wedge (w_{33} > 0) \implies w_{12} + w_{22} > 0 \quad \text{Contradiction}$$

$$(w_{12} \leq 0) \wedge (w_{13} \leq 0) \wedge (w_{23} \leq 0) \wedge (w_{32} \leq 0) \implies w_{13} + w_{23} \leq 0 \quad \text{Contradiction}$$

- 2nd case: $(w_{22} \leq 0) \wedge (w_{33} \leq 0) \wedge (w_{23} > 0) \wedge (w_{32} > 0)$

$$(w_{12} > 0) \wedge (w_{13} > 0) \wedge (w_{23} > 0) \wedge (w_{32} > 0) \implies w_{12} + w_{32} > 0 \quad \text{Contradiction}$$

$$(w_{12} \leq 0) \wedge (w_{13} \leq 0) \wedge (w_{22} \leq 0) \wedge (w_{33} \leq 0) \implies w_{13} + w_{33} \leq 0 \quad \text{Contradiction}$$

Hence

$$\text{sign}(w_{12} + w_{22}) = \text{sign}(w_{13} + w_{33}) = 1 - \text{sign}(w_{12} + w_{32}) = 1 - \text{sign}(w_{13} + w_{23}) \quad (9.4)$$

Specifically, if we apply the SBN Constraint

$$w_{3j} \leq 0 \quad \forall j \in \llbracket 1, 2 \rrbracket$$

from 8.13 deduced in Chapter 8, it leads to the following scenario (illustrated in Figure 9.2)

$$(\text{sign}(w_{23}) = \text{sign}(w_{32}) = 0) \wedge (\text{sign}(w_{22}) = \text{sign}(w_{33}) = 1)$$

Consequently, we can deduce the following constraints:

$\text{sign}(w_{12})$	$\text{sign}(w_{13})$	1	1	0	0
$\text{sign}(w_{22})$	$\text{sign}(w_{23})$	1	0	1	0
$\text{sign}(w_{32})$	$\text{sign}(w_{33})$	0	1	0	1

Figure 9.2: Representation of matrix $\text{sign}(W)$ applying SBN Constraints

- $w_{22} > 0, w_{23} \leq 0, w_{32} \leq 0, w_{33} > 0, w_{12} > 0, w_{13} > 0$: By definition, we have

$$\begin{cases} w_{12} + w_{22} > 0 \\ w_{13} + w_{33} > 0 \end{cases} \implies \begin{cases} w_{12} + w_{32} \leq 0 \\ w_{13} + w_{23} \leq 0 \end{cases} \implies \begin{cases} 0 < w_{12} \leq |w_{32}| \\ 0 < w_{13} \leq |w_{23}| \end{cases}$$

Assume

$$\begin{cases} w_{22} + w_{32} > 0 \\ w_{23} + w_{33} > 0 \end{cases} \implies \begin{cases} w_{22} > |w_{32}| \\ w_{33} > |w_{23}| \end{cases}$$

Then

$$\begin{cases} w_{12} + w_{22} + w_{32} > 0 \\ w_{13} + w_{23} + w_{33} > 0 \end{cases}$$

Assume that

$$\begin{cases} w_{22} + w_{32} \leq 0 \\ w_{23} + w_{33} \leq 0 \end{cases}$$

Then

$$\begin{aligned} & \begin{cases} 0 < w_{22} \leq |w_{32}| \\ 0 < w_{33} \leq |w_{23}| \end{cases} \\ \implies & \begin{cases} 0 < w_{12} + w_{22} \leq 2|w_{32}| \\ 0 < w_{13} + w_{33} \leq 2|w_{23}| \end{cases} \\ \implies & \begin{cases} w_{32} < w_{12} + w_{22} + w_{32} \leq |w_{32}| \\ w_{23} < w_{13} + w_{23} + w_{33} \leq |w_{23}| \end{cases} \end{aligned}$$

- $w_{22} > 0, w_{23} \leq 0, w_{32} \leq 0, w_{33} > 0, w_{12} \leq 0, w_{13} \leq 0$: By definition

$$\begin{cases} w_{12} + w_{32} \leq 0 \\ w_{13} + w_{33} \leq 0 \end{cases} \implies \begin{cases} w_{12} + w_{22} > 0 \\ w_{13} + w_{33} > 0 \end{cases} \implies \begin{cases} w_{22} > |w_{12}| \\ w_{33} > |w_{13}| \end{cases}$$

Assume that

$$\begin{cases} w_{22} + w_{32} > 0 \\ w_{23} + w_{33} > 0 \end{cases} \implies \begin{cases} w_{22} > |w_{32}| \\ w_{33} > |w_{23}| \end{cases}$$

Therefore

$$\begin{cases} w_{12} + 2w_{22} + w_{32} > 0 \\ w_{13} + 2w_{23} + w_{33} > 0 \end{cases}$$

Assume that

$$\begin{cases} w_{22} + w_{32} \leq 0 \\ w_{23} + w_{33} \leq 0 \end{cases} \implies \begin{cases} 0 < w_{22} \leq |w_{32}| \\ 0 < w_{33} \leq |w_{23}| \end{cases}$$

Therefore

$$\begin{cases} w_{12} + w_{22} + w_{32} \leq 0 \\ w_{13} + w_{23} + w_{33} \leq 0 \end{cases}$$

9.2 Closure constraints 4D

Similar as 3D, the definition of closure implies that: $\forall (x_1, x_2) \in \mathcal{B}^2, k = k_1 \oplus k_2$,

$$\text{sign} \circ f_3(\mathbf{x}^E) + (2k - 1)\text{sign} \circ f_4(\mathbf{x}^E) = k$$

$$\Leftrightarrow \begin{cases} f_3 \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{pmatrix} \right) - \text{sign} \circ f_4 \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \\ 0 \end{pmatrix} \right) = f_3 \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \\ 1 \end{pmatrix} \right) - \text{sign} \circ f_4 \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \\ 1 \end{pmatrix} \right) = 0 \\ f_3 \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \\ 1 \end{pmatrix} \right) + \text{sign} \circ f_4 \left(\begin{pmatrix} x_1 \\ x_2 \\ 0 \\ 1 \end{pmatrix} \right) = f_3 \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \\ 0 \end{pmatrix} \right) + \text{sign} \circ f_4 \left(\begin{pmatrix} x_1 \\ x_2 \\ 1 \\ 0 \end{pmatrix} \right) = 1 \end{cases}$$

For every triplet (G^0, G^1, G) that satisfies the predicate, it must simultaneously fulfill the constraints above for all $k \in \mathcal{B}$. Consequently, we examine their intersection:

- $k = k_1 \oplus k_2 = 0$:

$$\begin{aligned} & (\text{sign}(w_{13}x_1 + w_{23}x_2) = \text{sign}(w_{14}x_1 + w_{24}x_2)) \\ & \wedge (\text{sign}(w_{13}x_1 + w_{23}x_2 + w_{33} + w_{34}) = \text{sign}(w_{14}x_1 + w_{24}x_2 + w_{34} + w_{44})) \end{aligned}$$

- $k = k_1 \oplus k_2 = 1$:

$$\begin{aligned} & (\text{sign}(w_{13}x_1 + w_{23}x_2 + w_{33}) + \text{sign}(w_{14}x_1 + w_{24}x_2 + w_{34}) = 1) \\ & \wedge (\text{sign}(w_{13}x_1 + w_{23}x_2 + w_{43}) + \text{sign}(w_{14}x_1 + w_{24}x_2 + w_{44}) = 1) \end{aligned}$$

Enumerating all possible values of x_1 to more precisely define the constraints:

- Let $x_1 = x_2 = 0$:

$$\begin{aligned} & (0 = 0) \wedge (\text{sign}(w_{33} + w_{34}) = \text{sign}(w_{43} + w_{44})) \\ & \wedge (\text{sign}(w_{33}) + \text{sign}(w_{34}) = 1) \wedge (\text{sign}(w_{43}) + \text{sign}(w_{44}) = 1) \end{aligned}$$

which is equivalent to

$$\begin{cases} ((w_{33} + w_{34} > 0) \wedge (w_{43} + w_{44} > 0)) \vee ((w_{33} + w_{34} \leq 0) \wedge (w_{43} + w_{44} \leq 0)) \\ ((w_{33} > 0) \wedge (w_{34} \leq 0)) \vee ((w_{33} \leq 0) \wedge (w_{34} > 0)) \\ ((w_{43} > 0) \wedge (w_{44} \leq 0)) \vee ((w_{43} \leq 0) \wedge (w_{44} > 0)) \end{cases}$$

We can deduce the following relationships through a proof by contradiction, using the same approach as in the 3D case:

$$\text{sign}(w_{33}) = \text{sign}(w_{44}) = 1 - \text{sign}(w_{43}) = 1 - \text{sign}(w_{34}) \quad (9.5)$$

- Let $x_1 = 1, x_2 = 0$:

$$\begin{aligned}
& (\text{sign}(w_{13}) = \text{sign}(w_{14})) \\
& \wedge (\text{sign}(w_{13} + w_{33} + w_{34}) = \text{sign}(w_{14} + w_{34} + w_{44})) \\
& \wedge (\text{sign}(w_{13} + w_{33}) + \text{sign}(w_{14} + w_{34}) = 1) \\
& \wedge (\text{sign}(w_{13} + w_{43}) + \text{sign}(w_{14} + w_{44}) = 1)
\end{aligned}$$

which leads to

$$\text{sign}(w_{13} + w_{33}) = \text{sign}(w_{14} + w_{44}) = 1 - \text{sign}(w_{13} + w_{34}) = 1 - \text{sign}(w_{14} + w_{43}) \quad (9.6)$$

using the same approach as in the proof of Constraint 9.4.

- Let $x_1 = 0, x_2 = 1$:

$$\begin{aligned}
& (\text{sign}(w_{23}) = \text{sign}(w_{24})) \\
& \wedge (\text{sign}(w_{23} + w_{33} + w_{34}) = \text{sign}(w_{24} + w_{34} + w_{44})) \\
& \wedge (\text{sign}(w_{23} + w_{33}) + \text{sign}(w_{24} + w_{34}) = 1) \\
& \wedge (\text{sign}(w_{23} + w_{43}) + \text{sign}(w_{24} + w_{44}) = 1)
\end{aligned}$$

which leads to

$$\text{sign}(w_{23} + w_{33}) = \text{sign}(w_{24} + w_{44}) = 1 - \text{sign}(w_{23} + w_{34}) = 1 - \text{sign}(w_{24} + w_{43}) \quad (9.7)$$

using the same approach as in the proof of Constraint 9.4.

- Let $x_1 = x_2 = 1$:

$$\begin{aligned}
& (\text{sign}(w_{13} + w_{23}) = \text{sign}(w_{14} + w_{24})) \\
& \wedge (\text{sign}(w_{13} + w_{23} + w_{33} + w_{34}) = \text{sign}(w_{14} + w_{24} + w_{34} + w_{44})) \\
& \wedge (\text{sign}(w_{13} + w_{23} + w_{33}) + \text{sign}(w_{14} + w_{24} + w_{34}) = 1) \\
& \wedge (\text{sign}(w_{13} + w_{23} + w_{43}) + \text{sign}(w_{14} + w_{24} + w_{44}) = 1)
\end{aligned}$$

which is equivalent to

$$\begin{aligned}
& \left\{ \begin{aligned} & ((w_{123} > 0) \wedge (w_{124} > 0)) \vee ((w_{123} \leq 0) \wedge (w_{124} \leq 0)) \\ & ((w_{123} + w_{33} + w_{34} > 0) \wedge (w_{124} + w_{43} + w_{44} > 0)) \\ & \vee ((w_{123} + w_{33} + w_{34} \leq 0) \wedge (w_{124} + w_{43} + w_{44} \leq 0)) \\ & ((w_{123} + w_{33} > 0) \wedge (w_{124} + w_{34} \leq 0)) \vee ((w_{123} + w_{33} \leq 0) \wedge (w_{124} + w_{34} > 0)) \\ & ((w_{123} + w_{43} > 0) \wedge (w_{124} + w_{44} \leq 0)) \vee ((w_{123} + w_{43} \leq 0) \wedge (w_{124} + w_{44} > 0)) \end{aligned} \right. \\
& \text{with } \left\{ \begin{aligned} & w_{123} = w_{13} + w_{23} \\ & w_{124} = w_{14} + w_{24} \end{aligned} \right.
\end{aligned}$$

which leads to

$$\begin{aligned}
\text{sign}(w_{123} + w_{33}) &= \text{sign}(w_{124} + w_{44}) = 1 - \text{sign}(w_{123} + w_{34}) = 1 - \text{sign}(w_{124} + w_{43}) \\
\text{sign}\left(\sum_{i=1}^3 w_{i3}\right) &= \text{sign}\left(\sum_{i=1}^2 w_{i4} + w_{44}\right) = 1 - \text{sign}\left(\sum_{i=1}^2 w_{i4} + w_{34}\right) = 1 - \text{sign}\left(\sum_{i=1}^3 w_{i4}\right)
\end{aligned} \tag{9.8}$$

using the same approach as in the proof of Constraint 9.4.

9.3 Generalized Closure Constraints

Hypothesis 9.1 (Generalized Closure Constraints for 2 control bits). *From the analysis of both 3D and 4D cases, we can deduce the following generalized closure constraints for 2 control bits:*

$$\begin{aligned}
&\text{sign}\left(\sum_{l=1, l \neq i, l \neq j}^n w_{li}x_l + w_{ii}\right) = \text{sign}\left(\sum_{l=1, l \neq i, l \neq j}^n w_{lj}x_l + w_{jj}\right) = \\
&1 - \text{sign}\left(\sum_{l=1, l \neq i, l \neq j}^n w_{li}x_l + w_{ij}\right) = 1 - \text{sign}\left(\sum_{l=1, l \neq i, l \neq j}^n w_{lj}x_l + w_{ji}\right)
\end{aligned}$$

These constraints apply to all vectors $\mathbf{x}^{\mathbf{E}} \in \mathcal{B}^n \setminus \{x_i, x_j\}$ where x_i, x_j are the 2 control bits, $(i, j) \in \llbracket 1, n \rrbracket, i < j$.

Chapter 10

Properties of weight matrices

Using the same approach as in the case with one control bit, we can deduce specific properties of the weight matrices W^0 , W^1 and W by integrating the SBN Constraints 8.22 and Closure Constraints 9.1 discussed in Chapter 8 and 9, respectively.

Hypothesis 10.1 (Structure of matrix W for 2 control bits).

$$W = \begin{pmatrix} \begin{matrix} w_{11} & \cdots & w_{1(n-2)} \\ \vdots & \ddots & \vdots \\ w_{(n-2)1} & \cdots & w_{(n-2)(n-2)} \end{matrix} & \begin{matrix} w_{1(n-1)} \\ \vdots \\ w_{(n-2)(n-1)} \end{matrix} & \begin{matrix} w_{1n} \\ \vdots \\ w_{(n-2)n} \end{matrix} \\ \begin{matrix} w_{(n-1)1} & \cdots & w_{(n-1)(n-2)} & w_{(n-1)(n-1)} > 0 \\ w_{n1} \leq 0 & \cdots & w_{n(n-2)} \leq 0 & w_{n(n-1)} \leq 0 \end{matrix} & \begin{matrix} w_{(n-1)n} \leq 0 \\ w_{nn} > 0 \end{matrix} \end{pmatrix} = \begin{pmatrix} L_1 \\ \vdots \\ L_{n-1} \\ L_n \end{pmatrix}$$

where

$$W^0 \equiv \begin{pmatrix} \begin{matrix} w_{11} & \cdots & w_{1(n-2)} \\ \vdots & \ddots & \vdots \\ w_{(n-2)1} & \cdots & w_{(n-2)(n-2)} \end{matrix} \\ w_{(n-1)1} + w_{n1} & \cdots & w_{(n-1)(n-1)} + w_{n(n-1)} \end{pmatrix} = \begin{pmatrix} L_1 \\ \vdots \\ L_{n-2} \\ L_{n-1} + L_n \end{pmatrix} \Bigg|_{dim=n-1}$$

$$W^1 \equiv \begin{pmatrix} L_1 + L_n \\ \vdots \\ L_{n-2} + L_n \\ L_{n-1} \end{pmatrix} \Bigg|_{dim=n-1}$$

Part IV

Appendix

.1 All possible SBN GTs (0D-2D)

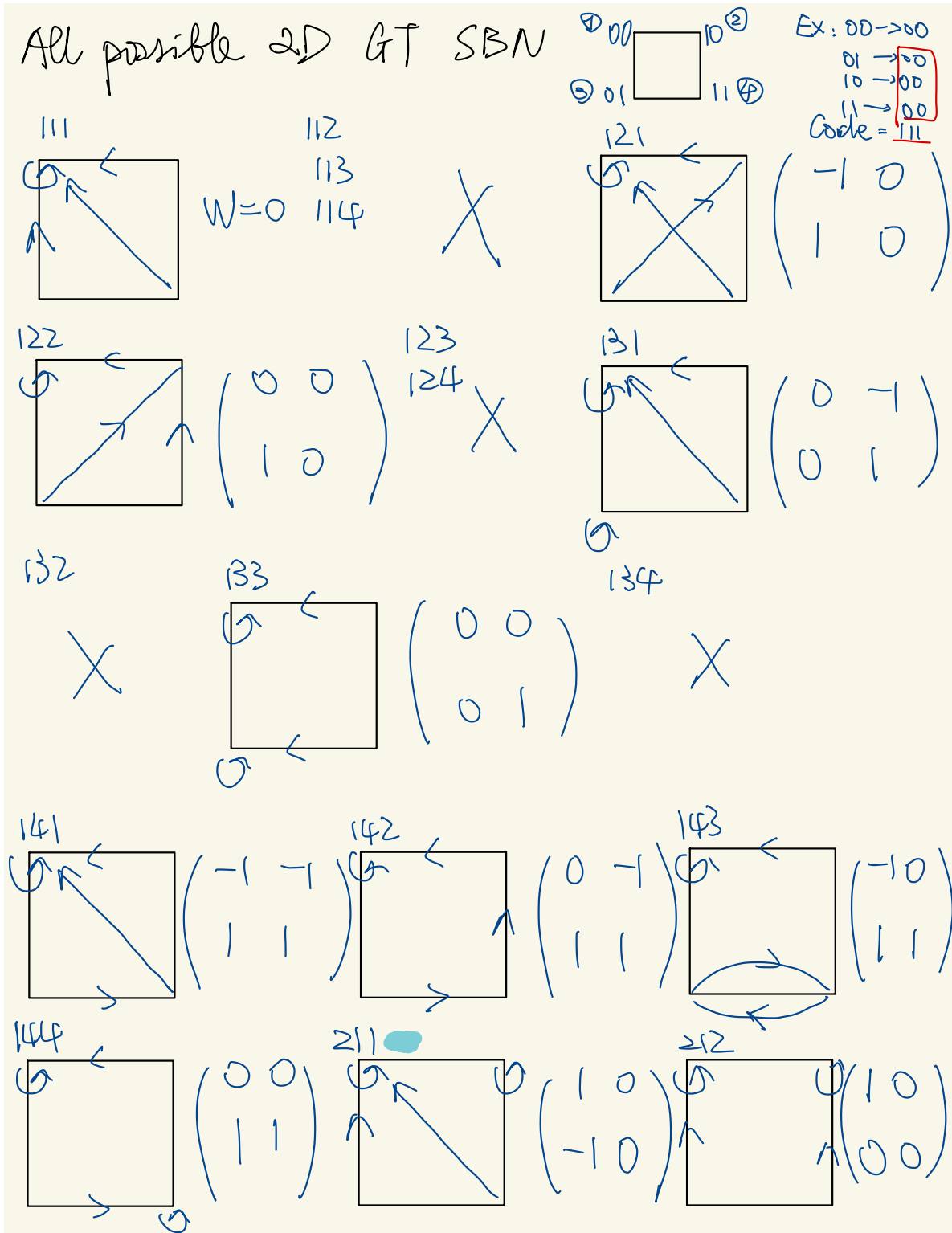


Figure 1: All possible TGs in 2D_1

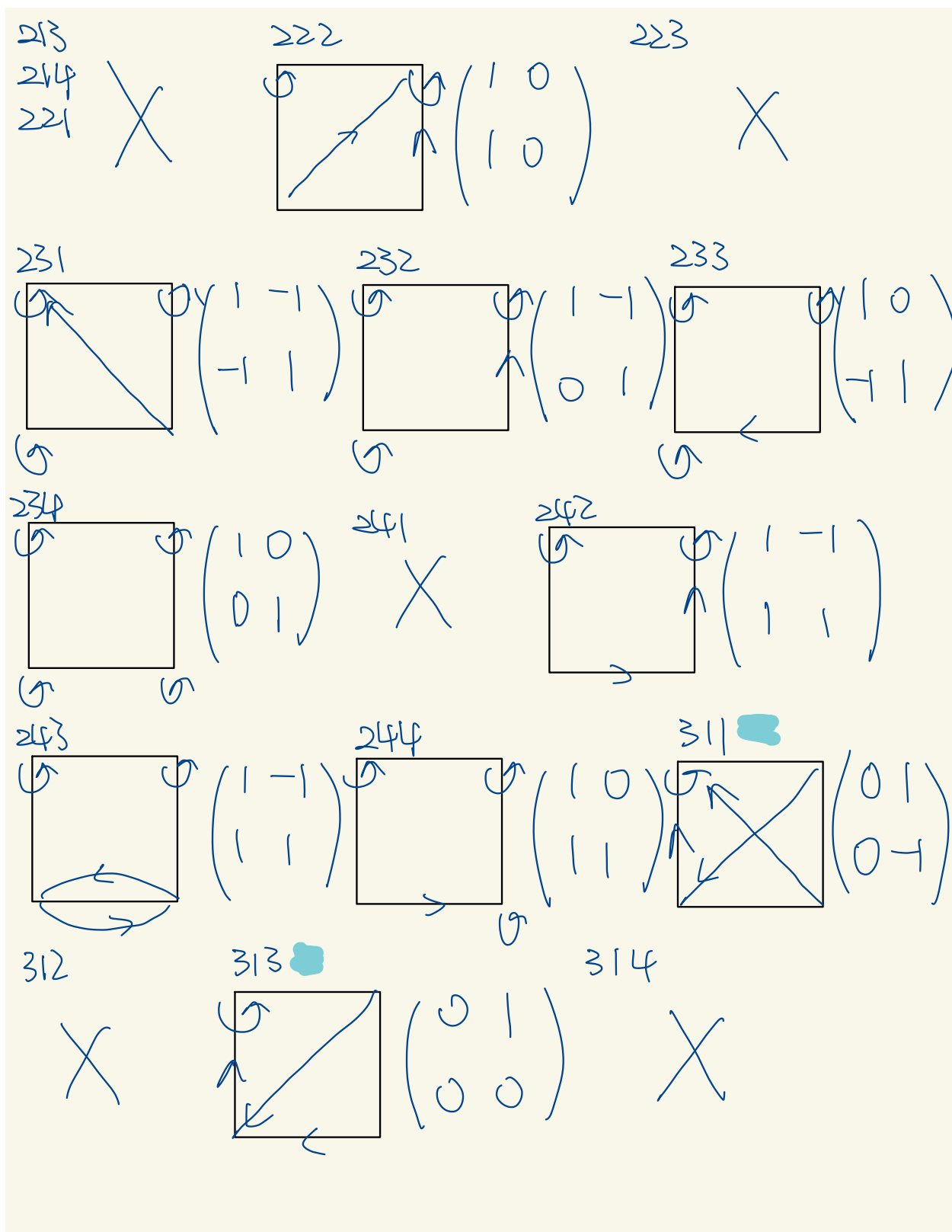


Figure 2: All possible TGs in 2D₂

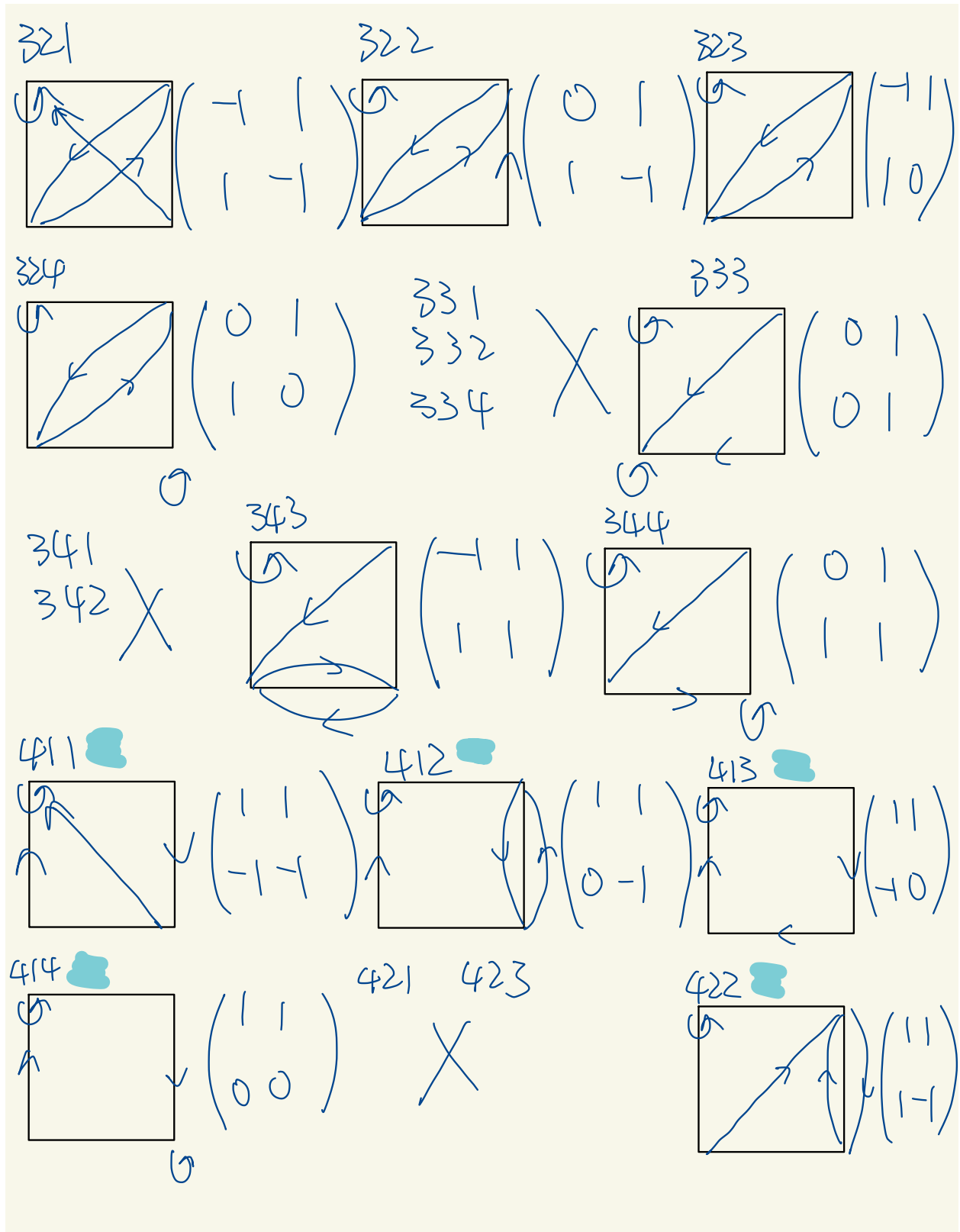
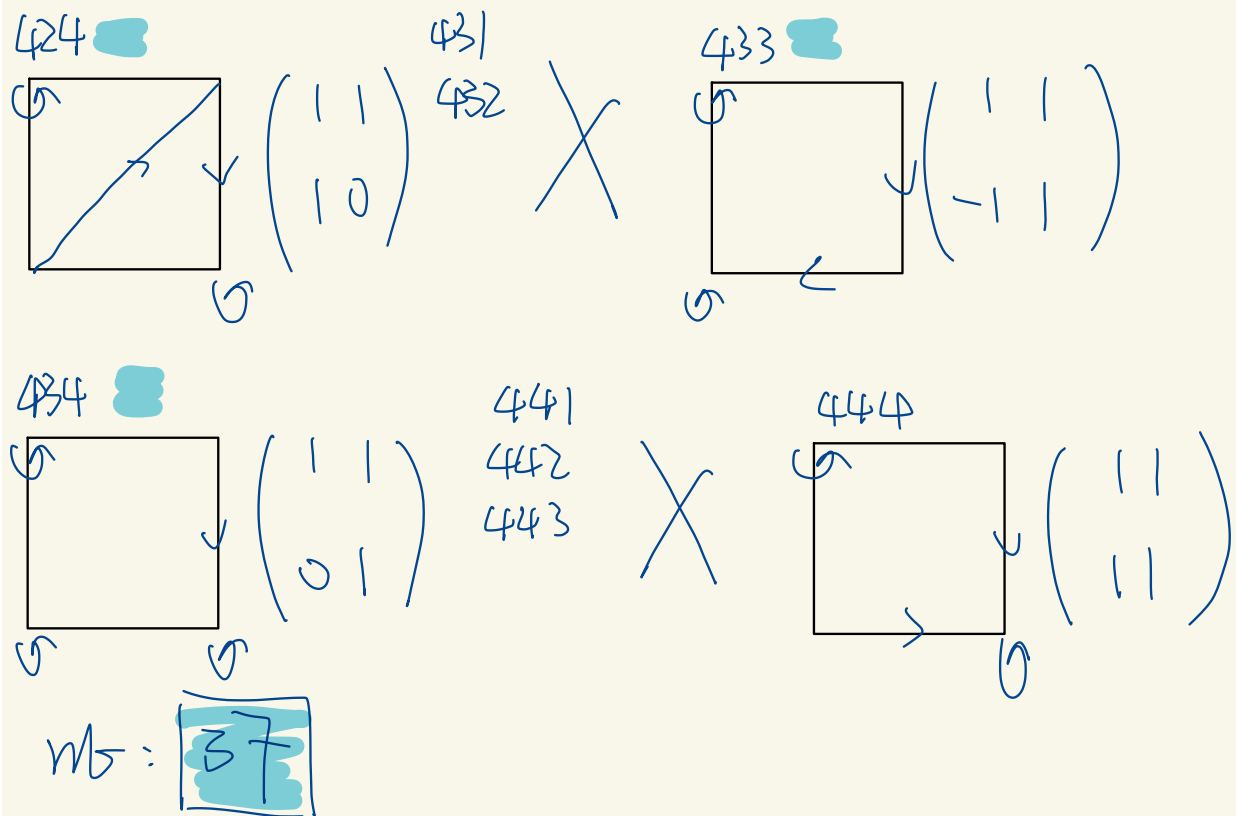
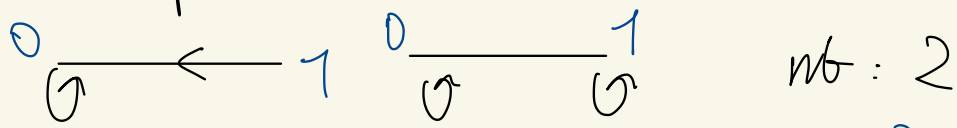


Figure 3: All possible TGs in 2D₃



All possible GT SBN 1D



nb: 2

All possible GT SBN 0D



nb: 1

Figure 4: All possible TGs in 2D_4